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Fisheries Research Report No. 346

West Coast Demersal Scalefish Resource Synopsis

Fisher, E.A., Fairclough, D.V.

2024

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Table of Contents

Executive Summary	1
List of Acronyms.....	2
1 Scope.....	3
2 Aquatic Environment of the West Coast Bioregion	4
3 Resource Description.....	6
3.1 Selection of Indicator Species	6
4 Species Description	9
4.1 Snapper.....	9
4.2 West Australian Dhufish	17
4.3 Baldchin Groper	23
4.4 Redthroat Emperor.....	27
4.5 Bight Redfish.....	31
4.6 Breaksea Cod.....	35
4.7 Hapuku	38
4.8 Bass Groper	41
4.9 Blue-Eye Trevalla.....	44
5 Fishery Information	47
5.1 Fisheries Capturing the West Coast Demersal Scalefish Resource	47
5.2 West Coast Demersal Scalefish (Interim) Managed Fishery	48
5.3 Recreational and Charter Fisheries.....	52
5.4 Customary Fishing	57
5.5 Illegal, Unreported or Unregulated Fishing	57
6 Fisheries Management.....	58
6.1 Management Framework.....	58
6.2 Harvest Strategy.....	59
6.3 External Influences	60
7 Information and Monitoring	61
7.1 Range of Information.....	61
7.2 Monitoring.....	62
7.3 Resource Assessments.....	65
References.....	69
Appendix 1. Productivity Susceptibility Analysis.....	83

Executive Summary

- The West Coast Demersal Scalefish Resource (WCDSR) comprises over 100 species that inhabit inshore (20-250 m deep) and offshore (>250 m deep) waters of the West Coast Bioregion (WCB; north of Kalbarri to east of Augusta).
- The WCB is predominantly a temperate oceanic zone with these waters influenced by the Leeuwin Current that transports warm, low-nutrient water southward along the western edge of the continental shelf.
- To monitor and assess the WCDSR, indicator species have been selected using a risk-based approach that considers their inherent vulnerability to fishing, level of targeting and catch taken by fishers, and the amount of information needed to manage stocks.
- The indicator species for inshore waters include Snapper, WA dhufish and, for the Mid-West area only, Baldchin groper, while indicators for offshore waters include Hapuku, Bass groper and Blue-eye trevalla.
- The WCDSR is primarily targeted by commercial, charter and recreational boat-based line fishers, with minor catches also landed by the commercial demersal gillnet and longline fisheries and the Western rock lobster fishery.
- Periodic assessments of key WCDSR indicator species have shown that heavy fishing pressure on the resource in the 1990s and 2000s resulted in depletion of stocks to below the limit reference level.
- After substantial management reform between 2007 and 2010, the WCDSR has been managed under a 20-year recovery plan designed to recover stocks.
- Refinements of the strategies used to monitor the recovery of this resource over the last few decades have improved the collection of biological and fishery data, enabling the progression to more sophisticated and dynamic models for assessing stock status of key species.
- A formal harvest strategy for the WCDSR was developed in 2021, with agreement by all sectors to explicit recovery objectives and control rules to track the rebuilding of stocks.
- This document provides a synopsis of current information about key species within the WCDSR, and the fisheries that target them. It is intended to be read in conjunction with published periodic stock assessments undertaken to monitor the recovery of this resource.

List of Acronyms

CDR	Catch disposal records
CSLPMF	Cockburn Sound Line and Pot Managed Fishery
ENSO	El Niño-Southern Oscillation
FHPA	Fish Habitat Protection Area
FRMA	<i>Fish Resources Management Act 1994</i>
GAB	Great Australian Bight
GVP	Gross value of production
IMCRA	Integrated Marine and Coastal Regionalisation of Australia
MLL	Minimum legal length
MTE	Metabolic Theory of Ecology
PRM	Post-release mortality
RFBL	Recreational Fishing from a Boat Licence
SNP	Single nucleotide polymorphisms
TDGDLF	Temperate Demersal Gillnet and Demersal Longline Fisheries
TL	Total length
VFAS	Voluntary Fisheries Adjustment Scheme
WA	Western Australia
WCB	West Coast Bioregion
WCDSIMF	West Coast Demersal Scalefish (Interim) Managed Fishery
WCDSR	West Coast Demersal Scalefish Resource
WCRLMF	West Coast Rock Lobster Managed Fishery
WDTF	Western Deepwater Trawl Fishery

1 Scope

This document provides a description of the West Coast Demersal Scalefish Resource (WCDSR) and the fishing activities (i.e. sectors/fisheries) affecting this resource in Western Australia (WA). The synopsis is focused on the six key species of demersal scalefish captured in the West Coast Bioregion (WCB) – Snapper, WA dhufish, Baldchin groper, Redthroat emperor, Bight redfish and Breaksea cod. Collectively, these species comprise more than 85% of retained catches by commercial, charter and recreational fisheries combined. The report also includes available information on three of the offshore demersal scalefish species caught in the WCB – Hapuku, Bass groper and Blue-eye trevalla.

This report outlines the range of information and data collected to monitor stocks of key species. It also provides a brief overview of the methods and models applied to assess stock status and determine the current risk to the resource. It is intended that the document is read in conjunction with published periodic stock assessments undertaken to track the recovery of this resource.

2 Aquatic Environment of the West Coast Bioregion

The marine environment of the WCB, extending over approximately 900 km of coastline from north of Kalbarri to east of Augusta (Figure 2.1), is predominantly a temperate oceanic zone with coastal water temperatures ranging around 18-24°C. These waters are heavily influenced by the Leeuwin Current that transports warm, low-nutrient water southward along the western edge of the continental shelf. The Leeuwin Current is strongest during the winter months, with annual variability in flow related to El Niño-Southern Oscillation (ENSO) events in the Pacific Ocean (Feng et al. 2003). In contrast, the Capes Current is generated off Cape Mentelle in the south-west (~ 34°S) by southerly winds in summer and transports cooler, nutrient-rich waters northwards along the inner shelf (Pearce and Pattiaratchi 1999).

The WCB comprises three distinct meso-scale bioregions classified by the Integrated Marine and Coastal Regionalisation of Australia (IMCRA 2006; Figure 2.1). The Abrolhos Islands region is in the northern section of the WCB around the Houtman Abrolhos Islands, approximately 60 km off the coast of Geraldton (Figure 3.1). This complex of 122 low-lying islands and reefs is well-known for its high diversity and unique mix of tropical and temperate species, recognised as the first Fish Habitat Protection Area (FHPA) in WA. Extensive habitat mapping of the shallow and deeper-water regions of the Abrolhos has been undertaken since the 1980s (e.g. Hatcher et al. 1988; Webster et al. 2002; Radford et al. 2008; Evans et al. 2012).

The Central West Coast region extends from the northern boundary of the WCB (27°S) to Perth (32°S), excluding the area around the Houtman Abrolhos Islands (Figure 3.1). This region is characterised by exposed sandy beaches and a limestone reef system that creates surface reef lines about 5 km from the coast. Further offshore, the continental shelf habitats are typically composed of coarse sand interspersed with low limestone reef associated with old shorelines. Surveys of deeper-water (35-75 m deep) habitats off Dongara, Jurien Bay and Lancelin have shown that the biota in the north is dominated by sponges, gradually transitioning to macroalgae-dominated habitats to the south (Bellchambers 2010).

The Leeuwin-Naturaliste region extends south from Perth to Black Point (115° 30'E), southeast of Augusta on the southern coast of WA. This is a transitional region between west and south coast communities, which is known for high species diversity and endemism. A significant impact of the clear and warm waters of the Leeuwin Current on this area is the growth and distribution of temperate seagrasses, with extensive seagrass meadows growing in the two significant marine embayments within this area; Cockburn Sound and Geographe Bay. Southward of Cape Naturaliste the coastline changes from limestone to predominantly granite as it becomes more exposed to the Southern Ocean.

The west coast of WA has been identified as a global warming hotspot (Hobday and Pecl 2014), having experienced marine heatwaves that have resulted in mortalities and shifting species distributions (Wernberg et al. 2013; Caputi et al. 2015; 2016).

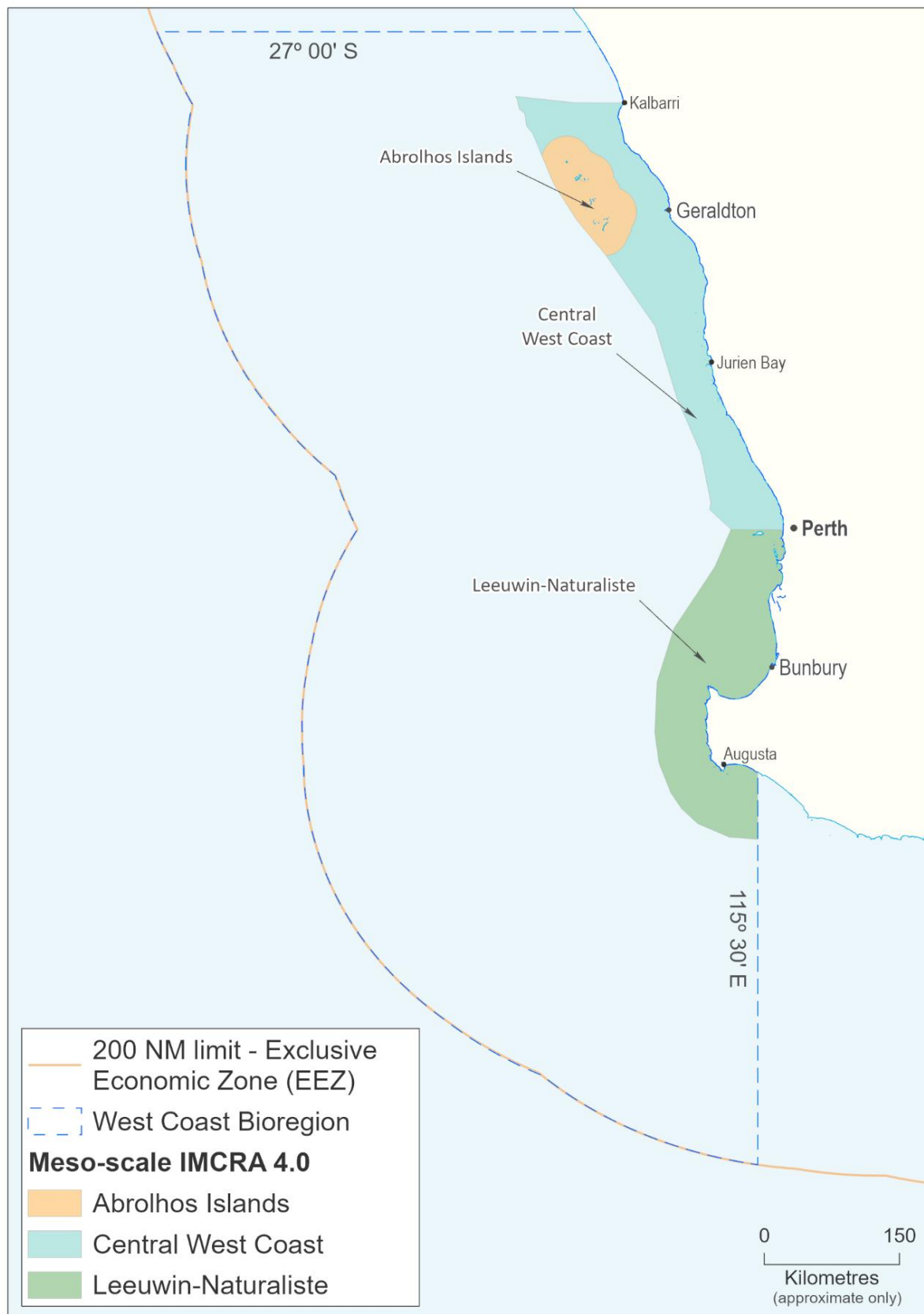


Figure 2.1. Map showing the three main IMCRA regions within the WCB.

3 Resource Description

The WCDSR comprises more than 100 species of demersal scalefish that are targeted in varying proportions by commercial, charter and recreational fishers in inshore and offshore waters of the WCB.

The inshore demersal suite is found in waters 20-250 m deep and collectively contributes more than 95% to demersal catches in the WCB (Table 3.1). Catches of inshore demersal scalefish in this region are dominated by six key species, all of which are caught commercially and recreationally (Table 3.1). WA dhufish, Baldchin groper and Breaksea cod are predominantly caught by recreational fishers, whereas Snapper, Redthroat emperor and Bight redfish are mostly caught by commercial fishers (Table 3.1).

The offshore demersal suite, which mostly occurs in waters >250 m deep, accounts for less than 5% of overall demersal scalefish catches in the WCB (Table 3.1). Catches are dominated by Hapuku and Greybanded grouper (formerly Eightbar grouper; Moore et al. 2022), with smaller catches of Blue-eye trevalla and Bass groper also retained primarily by recreational fishers (Table 3.1).

3.1 Selection of Indicator Species

The process for monitoring and assessing the finfish resources of WA involves allocating the species within each bioregion into one of five suites – Estuarine, Nearshore, Inshore Demersal, Offshore Demersal and Pelagic (Department of Fisheries 2011). For each bioregion, a risk-based approach has been used to identify indicator species from the list of species within each suite, based on their inherent vulnerability to fishing and other considerations such as their economic and social values (Lenanton et al. 2006; Newman et al. 2018). The status of these indicator species is assumed to represent the status of the entire suite.

The indicator species selected for assessing the status of the inshore suite within the overall WCB are Snapper and WA dhufish (Lenanton et al. 2006; Newman et al. 2018). Baldchin groper is an indicator species for the Mid-West area of the WCB only (DPIRD 2021). The high indicator scores for these three species are driven by their high inherent vulnerability (relating to longevity of WA dhufish and Snapper, protogynous sex change in Baldchin groper, and restricted distributions of the endemic WA dhufish and Baldchin groper), the high fishing pressure on these stocks (collectively comprising around 75% of the total demersal scalefish catch in the WCB; Table 3.1), and high management importance resulting from their value to fishers (Newman et al. 2018; Table 3.2).

The indicator species initially selected for the offshore suite in the WCB included Hapuku, Greybanded grouper and Blue-eye trevalla (Lenanton et al. 2006). These were recently revised after new biological information for Greybanded grouper indicated no spawning within temperate waters of the WCB (Wakefield et al. 2013a). Greybanded grouper was replaced as an offshore indicator by Bass groper as there has been a recent increase in recreational catches of this species (Ryan et al. 2019; 2022; Table 3.2).

Table 3.1. Retained catches (tonnes, t) of demersal scalefish landed by commercial, charter and recreational fishers in the WCB in 2021 (commercial and charter) or 2020-21 (recreational; see Ryan et al. 2022). Note that catches have been rounded and may not add up to totals.

Species	Commercial	Charter*	Recreational*	Total
Total demersal	253	54	228	535
Total inshore	244	54	217	515
Snapper	109	25	52	186
WA dhufish	41	10	108	158
Baldchin groper	9	11	38	58
Redthroat emperor	27	2		29
Bight redfish	23	0.2		23
Breaksea cod	2	2	10	14
Spangled emperor	8	3		11
Blue morwong	2	1	4	7
Western blue groper	6			6
Goldband snapper	4			4
Emperors (other)	1	0.3	2**	3
Red emperor	2			2
Northern pearl perch	2			2
Rockcods	2			2
Redfishes (other)	0.2	0.2	1***	2
Grunter breams	2			2
Foxfish	0.1	0.2	1	1
Rankin cod	1			1
Painted sweetlips	1			1
Boarfishes	1			1
Sergeant baker	0.04	0.1	1	1
Sea sweep	0.004	0.1	1	1
Other inshore species	2			2
Total offshore	9	0.3	11	20
Hapuku	4	0	3	7
Greybanded grouper	4	0.2	3	7
Blue-eye trevalla	1	0.02	3	4
Bass groper	0.1	0.1	2	3
Ruby snapper	0	0	0.3	0.3

* Top 15 species/groups

** All emperor spp., including Redthroat emperor, Spangled emperor

*** All redfish spp., including Bight redfish

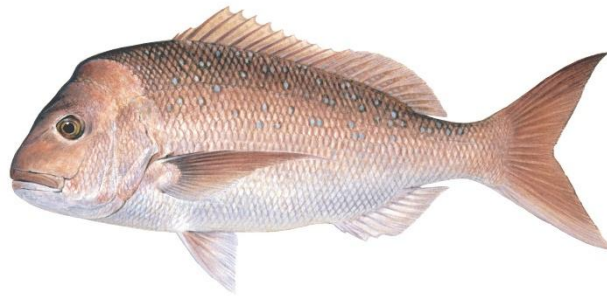
Table 3.2. Risk scores informing the selection of indicator species for the inshore and offshore suites of the WCDSR (see Newman et al. 2018 for description of methodology).

Species	Inherent vulnerability	Risk to wild stock	Management importance	Indicator Value
Inshore suite				
WA dhufish	4	4	3.2	51.2
Snapper	3	4	3.4	40.8
Baldchin groper	4	4	2.4	38.4
Western blue groper	4	3	2.2	26.4
Redthroat emperor	4	4	1.6	25.6
Bight redfish	4	4	1.6	25.6
Coral trout	4	3	1.6	19.2
Swallowtail	4	4	0.8	12.8
Yelloweye redfish	4	4	0.8	12.8
Breaksea cod	3	3	1.4	12.6
Foxfish	4	3	1.0	12.0
Spangled emperor	3	3	1.2	10.8
Sergeant baker	3	3	1.2	10.8
Blue morwong	3	3	0.8	7.2
Sea sweep	4	3	0.6	7.2
Harlequin fish	3	3	0.6	5.4
Offshore suite				
Hapuku	4	3	1.0	12.0
Bass groper	4	3	0.8	9.6
Blue-eye trevalla	4	3	0.8	9.6
Greybanded grouper	4	3	0.6	7.2
Ruby snapper	4	3	0.6	7.2

4 Species Description

This section summarises available biological information for the key WCDSR species on which this report is focused, including Snapper, WA dhufish, Baldchin groper, Redthroat Emperor, Bight redfish, Breaksea cod, Hapuku, Bass groper and Blue-eye trevalla. The productivity of these species is also summarised in Appendix 1.

4.1 Snapper



4.1.1 Taxonomy and Distribution

Snapper (*Chrysophrys auratus*), commonly referred to as pink snapper, is a member of the family Sparidae (sea breams). Previously known as *Pagrus auratus*, this species is characterised by pale pink to dark red upper body colouration and dorsal surfaces scattered with small, bright blue spots that become less distinct with age (Hutchins and Swainston 1999). Large mature fish develop a large bony hump on the head, which is more prominent in males.

Snapper are widely distributed throughout the warm temperate and subtropical waters of the western Indo-Pacific, around southern Australia and New Zealand (Gomon et al. 2018). In Australia they are distributed from Barrow Island in WA, around the south of the continent, to Hinchinbrook Island in northern Queensland (Kailola et al. 1993).

4.1.2 Stock Structure

The stock structure of Snapper in WA is complex. Three genetically related but biologically separate stocks have been identified within the inner gulfs of Shark Bay, based on evidence from genetic analyses (Johnson et al. 1986; Gardner et al. 2017), otolith microchemistry (Edmonds et al. 1999; Bastow et al. 2002), hydrodynamic modelling of egg and larval dispersal (Nahas et al. 2003), and biological and tagging studies (Moran et al. 2003; Jackson 2007; Norriss et al. 2012). In contrast, Snapper in oceanic waters off the WA coast show lower levels of genetic differentiation over hundreds of kilometres. Although both analysis of microsatellites and single nucleotide polymorphisms (SNPs) identified isolation by distance along the west coast, the latter approach also found genetic discontinuities between the mid-west (Gascoyne coast to Lancelin), south-west (Cockburn Sound to Albany) and south-east (Esperance to Ceduna) (Bertram et al. 2022; Gardner et al. 2022), with the boundaries of these three stocks likely shaped by oceanography and coastal topography, as well as spawning time and location (Bertram et al. 2022).

Analyses of otolith microchemistry of Snapper along the WA coast indicate residency of adults at the scale of management areas (within the WCB), with recruitment likely coming from multiple nursery areas (Fairclough et al. 2013; Jackson et al. 2023). These findings are supported by tagging studies, which show that the majority of adults tagged at key spawning locations are recaptured within 100 km and also provide evidence of philopatry of individuals to spawning aggregations in the WCB (Moran et al. 2003; Wakefield et al. 2011; Crisafulli et al. 2019). While eddies help facilitate the retention of Snapper eggs and larvae in key spawning areas, spawning also occurs along the open coastline and therefore, some early life history stages are transported by ocean currents along more open waters of the west coast of WA (Nahas et al. 2003; Wakefield 2010; Breheny et al. 2012; Jackson et al. 2023).

Current assessments of Snapper in the WCB provide separate estimates of stock status for the northern (Kalbarri and Mid-West) and southern (Metropolitan and South-West) management areas, of which the boundaries broadly align with those delineating the mid-west and south-west stocks identified by the genomic study (Bertram et al. 2022). While hydrodynamic modelling has shown that the southward-flowing Leeuwin Current can facilitate some north to south dispersal of eggs and larvae from the Gascoyne to Kalbarri, and Kalbarri to Mid-West, the majority recruit locally, supported by the greater genomic relatedness of adults and juveniles in each management area than to adjacent areas (Jackson et al. 2023).

4.1.3 Life History

The sub-sections below provide an overview of life history information for Snapper in the WCB, with a summary of the relevant biological parameters for the northern (Kalbarri and Mid-West) and southern (Metropolitan and South-West) management areas presented in Table 4.1 and Table 4.2, respectively.

4.1.3.1 Habitats and Movements

Juvenile Snapper inhabit sheltered inshore waters such as embayments and inlets, typically remaining in such nursery areas for the first two years of life before migrating offshore (Kailola et al. 1993; Wakefield et al. 2011). On the lower west coast of WA, the lengths and ages of immature individuals progressively increase with an increased distance from the key spawning and nursery areas of Cockburn Sound, Warnbro Sound and Owen Anchorage (Wakefield et al. 2011). Sub-adult and adult Snapper typically associate with rocky reefs in offshore waters up to 200 m deep (Kailola et al. 1993).

Studies of Snapper in Australia and New Zealand suggest that the majority of individuals are essentially resident to an area and undertake only localised movements associated with feeding and spawning, with a small number of older juveniles and adults found to move larger distances (Sanders 1973; Moran et al. 2003; Wakefield et al. 2011; Crisafulli et al. 2019). In the WCB, tagging studies show seasonal migrations of adults in the Metropolitan area to spawn in Cockburn Sound (Wakefield et al. 2011; Crisafulli et al. 2019).

Table 4.1. Summary of biological parameters for Snapper in the Kalbarri and Mid-West areas of the WCB. F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters (Schnute)		$L_t = \left((l_1^b + (l_2^b - l_1^b)) \frac{1 - \exp(-a(t - t_1))}{1 - \exp(-a(t_2 - t_1))} \right)^{1/b}$ Estimated using age- and length-based catch curve model in L3Assess (Hesp 2023), where l_1 and t_1 were set to 0 and t_2 was set to 20
l_2 (mm TL)	784.1 (F), 764.3 (M)	Updated 2024
a	0.126 (F), 0.131 (M)	Updated 2024
b	1.699 (F), 1.753 (M)	Updated 2024
Maximum age (years)	30	DPIRD biological data
Maximum size (mm TL)	1,005	DPIRD biological data
Natural mortality, M (y^{-1})	0.14	Hewitt and Hoenig 2005, based on max age of 30 years
Length-weight parameters		W (g) = a TL (mm) ^{b}
a	9.89×10^{-6}	Updated 2023
b	2.669	Updated 2023
Reproduction	Non-functional hermaphrodite, multiple (batch) spawner, indeterminate fecundity	Wakefield 2010, Jackson et al. 2012
Maturity parameters		Logistic
A_{50} (years)	5.5 (F), 5.2 (M)	Fairclough et al. 2021
A_{95} (years)	14.9 (F), 13.6 (M)	Fairclough et al. 2021
L_{50} (mm TL)	523 (F)	Fairclough et al. 2021
L_{95} (mm TL)	843 (F)	Fairclough et al. 2021
Maximum batch fecundity	Up to 660,000 eggs per batch	Mackie et al. 2009
Fecundity parameters (batch)		$BF = a FL^b$
a	9.436×10^{-5}	Jackson et al. 2012
b	3.359	Jackson et al. 2012

Table 4.2. Summary of biological parameters for Snapper in the Metropolitan and South-West areas of the WCB. F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters (Schnute)		$L_t = \left((l_1^b + (l_2^b - l_1^b)) \frac{1 - \exp(-a(t - t_1))}{1 - \exp(-a(t_2 - t_1))} \right)^{1/b}$ Estimated using age- and length-based catch curve model in L3Assess (Hesp 2023), where l_1 and t_1 were set to 0 and t_2 was set to 20
l_2 (mm TL)	927.2 (F), 915.9 (M)	Updated 2024
a	0.160 (F), 0.168 (M)	Updated 2024
b	1.239 (F), 1.206 (M)	Updated 2024
Maximum age (years)	40	Norriss and Crisafulli 2010
Maximum size (mm TL)	1,095	DPIRD biological data
Natural mortality, M (y^{-1})	0.12	Wise et al. 2007
Length-weight parameters		W (g) = a TL (mm) ^{b}
a	3.67×10^{-5}	Updated 2023
b	2.830	Updated 2023
Reproduction	Non-functional hermaphrodite, aggregate to spawn, multiple (batch) spawner, indeterminate fecundity	Wakefield 2010, Jackson et al. 2012
Maturity parameters		Logistic
A_{50} (years)	6.7 (F), 5.5 (M)	Fairclough et al. 2021
A_{95} (years)	12.9 (F), 12.3 (M)	Fairclough et al. 2021
L_{50} (mm TL)	660 (F)	Fairclough et al. 2021
L_{95} (mm TL)	900 (F)	Fairclough et al. 2021
Maximum batch fecundity	Up to 660,000 eggs per batch	Mackie et al. 2009
Fecundity parameters (batch)		$BF = a FL^b$
a	9.436×10^{-5}	Jackson et al. 2012
b	3.359	Jackson et al. 2012

4.1.3.2 Age and Growth

Snapper are relatively long-lived, with a maximum recorded age of 40 years in WA (Norris and Crisafulli 2010). The ages of Snapper are determined using counts of annuli in sectioned sagittal otoliths, with the periodicity of increment formation validated using marginal increment analyses (Wakefield et al. 2017). The period of deposition of the opaque zone each year varies among regions, occurring between May and September on the subtropical upper west coast (Gascoyne) and between August and December in the temperate lower west and south coasts of WA (Wakefield et al. 2017).

Growth studies of Snapper in the WCB have shown that estimates of von Bertalanffy growth parameters for females and males are not biologically different, however, parameter estimates differ markedly between regions (Lenanton et al. 2009; Fairclough et al. 2021; Wakefield et al. 2017). Snapper in the warmer waters of the northern management areas (Kalbarri and Mid-West) attain a smaller size than those in the southern management areas (Metropolitan and South-West) (Table 4.1 and Table 4.2). Recent growth analyses for Snapper in each management area have shown that the Schnute growth model provides statistically better fits than the von Bertalanffy growth model to available length-at-age data.

4.1.3.3 Natural Mortality

The value of natural mortality (M) of 0.12 y^{-1} used in the assessment of Snapper in the southern parts of the WCB was calculated by Wise et al. (2007) using two different empirical estimation methods (Hoenig 1983; Hewitt and Hoenig 2005), based on the previously recorded maximum age of 39 years for this species in WA. For Snapper in the northern WCB, where the maximum recorded age is 30 years, an updated M estimate of 0.14 y^{-1} has been derived from the equation described by Hewitt and Hoenig (2005) of $M = 4.22/A_{\text{max}}$, which corresponds closely to the assumption that 1.5% of fish survive to the observed maximum age (e.g., Dureuil and Froese 2021) (Table 4.1 and Table 4.2).

4.1.3.4 Reproduction

Snapper are non-functional hermaphrodites, with pre-maturational protogynous sex change (early bisexual juvenile stage but separate sexes thereafter) occurring in some individuals (Francis and Pankhurst 1988; Wakefield 2006, 2010; Sadovy de Mitcheson and Liu 2008). The mean sizes and ages at maturity of Snapper vary between regions of WA and is understood to be largely influenced by water temperature (Jackson 2007; Lenanton et al. 2009; Wakefield et al. 2015). Snapper in the northern parts of the WCB (Kalbarri and Mid-West) attain maturity at a smaller size and younger age than in the cooler waters of the southern WCB (Table 4.1 and Table 4.2).

Snapper spawn in aggregations in some areas of the WCB (see below), with synchronous spawning around the lunar cycle (Wakefield 2006; 2010). During spawning, pelagic eggs are released and fertilized within the upper layers of the water column (Jackson 2007). The egg incubation period is influenced by environmental conditions, particularly water temperature (Wakefield 2010). Following an initially pelagic larval phase of around 20-25 days, metamorphosis and settlement occurs over soft, muddy bottoms at approximately 9-12 mm in length (Kingsford and Atkinson 1994; Trnski 2002; Fowler and Jennings 2003).

The three adjacent embayments of Owen Anchorage, Cockburn Sound and Warnbro Sound are the only locations in the WCB where Snapper are known to aggregate in large numbers during the peak spawning period (Wakefield 2010). Eggs are retained in Cockburn Sound by a wind-driven gyre, where they settle as larvae and remain for the first two years. Spawning Snapper are also found within the oceanic waters of the WCB, providing greater potential for alongshore dispersal of eggs and larvae (Lenanton et al. 2009).

Snapper are batch spawners with indeterminate fecundity, capable of spawning multiple times during an extended spawning period. Based on data from the inner gulfs of Shark Bay, a positive relationship exists between batch fecundity and the size of females (Jackson et al. 2012). For females ranging from 172 to 741 mm FL, batches of between ~1,500 and 660,000 eggs have been estimated (Mackie et al. 2009). Spawning occurs predominantly at night during the three hours following the high tide (Wakefield 2010; Jackson et al. 2012). The female spawning fraction peaks at the new moon and, to a lesser extent, the full moon, coinciding with tidal flows that may contribute to greater dispersal of eggs and larvae (Wakefield 2010).

The timing and duration of the Snapper spawning season varies spatially and temporally, with water temperature regulating gonad development in this species (Scott and Pankhurst 1992). Spawning at each latitude peaks at 19-21°C, following a decline in temperature into winter in the subtropical Gascoyne region and after a rise in temperature in spring along the temperate south-west and south coasts (Wakefield et al. 2015). Spawning on the upper west coast of WA extends from May to October, with an assumed birth date of 1 August corresponding to the midpoint of this spawning period. On the lower west coast, the main spawning period extends from August to January, with a specified birth date of 1 November (Wakefield et al. 2015).

4.1.3.5 Factors Affecting Recruitment and Other Biological Parameters

Variations in the abundance of age cohorts and annual variability in catch rates of Snapper in the WCB suggest that the recruitment success of this species fluctuates substantially between years. Trawl surveys in Cockburn Sound have indicated two strong years of Snapper recruitment from 1999 and 2007 (Johnston et al. 2008), with the latter of these cohorts making a large contribution to catches in both the southern and northern management areas between 2009 and 2014 (Fairclough et al. 2021). Highly variable annual recruitment of Snapper has also been shown elsewhere in Australia and New Zealand (Francis 1993; Fowler and Jennings 2003; Hamer and Jenkins 2004). The reliance of a stock on years of exceptionally good recruitment is common to many long-lived species and has been termed the storage effect (Warner and Chesson 1985).

Although the driving mechanisms of recruitment variability in Snapper are not entirely understood, water temperature during, or shortly after, the spawning season has been found to be positively correlated with recruitment success (Francis 1993; Fowler and Jennings 2003; Parsons et al. 2014). Recruitment variation in this species has been linked to variability in larval prey densities (Zeldis et al. 2005; Murphy et al. 2012).

Research recently completed used genomics to further investigate the connectivity of Snapper within Cockburn Sound and adjacent regions of the lower west coast of WA,

from the northern metropolitan region to the south-west, to evaluate the importance of these spawning aggregations as a recruitment source for the broader stock (Fairclough et al. 2025). This information will inform future port developments in this area, given the potential for impacts of dredging and shipping in the area on the viability of spawning aggregations, as well as the survival and growth of eggs, larvae and juveniles (Partridge and Michael 2010; Wakefield et al. 2013b). The study demonstrated that adults in Cockburn Sound and other west coast spawning locations were more closely related to the juveniles at those locations than elsewhere, demonstrating localised recruitment. However, there were high genetic migration rates between locations. Together this indicates that spawning in each of the metropolitan embayments and along the coast is important for population replenishment. However, the adults that spawn in the metropolitan embayments are exclusively large fish and therefore can contribute relatively more to recruitment than individuals along the coast (Fairclough et al. 2025).

As the growth, spawning activity and early development of Snapper is strongly influenced by water temperature (Wakefield 2010; Wakefield et al. 2015), this species in the WCB has a medium sensitivity to climate change, i.e. gradual ocean warming and extreme events such as marine heatwaves (Caputi et al. 2015; Newman et al. 2024). The impacts of any temperature-related effects of climate change on this species are expected to vary across its range (Jackson et al. 2020; 2023). For example, as most spawning activity occurs within a relatively small temperature range (Wakefield 2006), increasing water temperatures could potentially lead to a latitudinal shift in key spawning grounds, reduced abundances in the northern areas and corresponding increases in areas to the south (Jackson et al. 2020). Any shift in spawning locations would also likely result in changes to Snapper recruitment patterns (Jackson et al. 2020).

4.1.3.6 Diet and Predators

Snapper are highly opportunistic carnivores with a broad ranging diet that can include a wide variety of mainly benthic organisms such as crustaceans, bivalves, cephalopods, marine worms, starfish, sea urchins and fish, depending on availability (Colman 1972; Usmar 2012). On the west coast of WA, dietary studies of this species have identified ontogenetic differences in diet relating to body size and season (French et al. 2012). With an increasing size, the diet shifted from predominantly ophiuroids to larger prey, such as brachyuran crabs, teleosts, echinoids and ultimately asteroids, likely reflecting a shift from foraging over soft sediments to reefs (French et al. 2012). Predators of demersal scalefish are likely to include larger teleosts such as cod (Serranidae), as well as sharks and dolphins. The trophic level of Snapper is 3.6 (Froese and Pauly 2023).

4.1.3.7 Parasites and Diseases

Snapper from offshore waters near Shark Bay have been found with a didymozoid trematode (*Gonapodasmius williamsoni*; Cribb and Williams 1992). The prevalence of this thread-like parasite in muscle tissue was greater in older and larger individuals, with few infections large enough to be noticed in fillets and unlikely pathogenic to humans (Williams et al. 1993). A large nematode (*Philometra* sp.) has been found to infect more than 50% of male testes and 11% of female ovaries of Snapper in New Zealand, with high prevalence of infestation by this parasite potentially impacting on reproductive output (Hine and Anderson 1981).

A betanodavirus, also known as nervous necrosis virus, has been detected in juvenile Snapper reared within a hatchery environment. This virus can cause viral encephalopathy and retinopathy (VER), a disease that affects more than 50 marine finfish species and can cause significant mortality impacting aquaculture production (Doan et al. 2017).

4.1.4 Inherent Vulnerability

The biological characteristics of Snapper make its stocks vulnerable to fishing in the absence of appropriate management (see Appendix 1). Although this species is relatively fast-growing and most individuals mature and become selected by fishers around 3-5 years of age, it has a potential longevity up to 40 years and highly variable recruitment. Whilst larger individuals are highly fecund, aggregations of spawning fish can be easily targeted unless protected by fishing closures. The selective removal by fishers of the larger and older fish in the population, that contribute substantially more to population replenishment than smaller, younger fish and would otherwise be able to spawn for multiple successive years, can severely reduce the resilience of stocks to fishing pressure (Longhurst 2002). For depleted stocks, newly recruited fish from a recent good recruitment event can be rapidly removed, which will impact strongly on stock recovery.

Snapper are primarily caught by line fishing close to reefs or other structure. Trends in recreational catch data suggest that a large proportion of Snapper are released due to being below the minimum legal length (MLL) (Ryan et al. 2021; Crisafulli et al. 2023). While based on experiments of relatively short-term effects on released fish, available information suggests that this species can be susceptible to post-release mortality when captured in deeper waters, or as a result of gut-hooking (StJohn et al. 2009; Broadhurst et al. 2012, Butcher et al. 2012; Maggs et al. 2024). On average, 25% of released Snapper in the WCB are expected to be susceptible to post-release mortality (DPIRD 2021).

4.2 West Australian Dhufish



4.2.1 Taxonomy and Distribution

The West Australian (WA) dhufish (*Glaucosoma hebraicum*) is a member of the monogeneric family Glaucosomatidae (pearl perches). This species is easily recognised by its silvery grey body with purple and bronze reflections, and a dark line through the eye that becomes less pronounced in larger fish (Hutchins and Swainston 1999). The dark, horizontal stripes characteristic of juvenile WA dhufish disappear as they mature and move onto reefs. Larger males can be distinguished by an elongate filament on the fourth soft dorsal ray (McKay 1997).

WA dhufish are endemic to the south-western coast of WA, between Shark Bay in the north and the Recherche Archipelago in the south (Hutchins and Swainston 1999). Within their range, this species is most abundant between the Houtman Abrolhos Islands and Cape Naturaliste (Lenanton et al. 2009).

4.2.2 Stock Structure

Molecular analyses of microsatellite DNA from WA dhufish have demonstrated low levels of genetic differentiation between individual management areas of the WCB, suggesting a single genetic stock across its range (Berry et al. 2012). Otolith microchemistry studies have demonstrated residency of adults within management areas, with recruitment of juveniles likely occurring from multiple adjacent nurseries within each area (Fairclough et al. 2013). The limited movement of this species is also supported by available tagging information (St John et al. 2009). As assemblages of WA dhufish are likely to be largely independent between management areas, there is potential for localised depletion and reduced reproductive capacity of this species in highly exploited, localised areas.

Hydrodynamic modelling provides further evidence that the majority of WA dhufish larvae recruit from within the management areas where they are spawned (Berry et al. 2012; Strzelecki et al. 2013). While the Capes Current and Leeuwin Current facilitate both norward and southward transport of larvae, dispersal patterns are likely dependent on the position on the shelf from which eggs are spawned and geographical features along the WA coast that influence retention rates (Feng et al. 2010; Pearce et al. 2011a). Particle tracking simulations indicate that the South-West management area has greater retention compared to other regions of the WCB, and the potential also for capturing long-distance recruits from the north (Berry et al. 2012).

4.2.3 Life History

The sub-sections below provide an overview of the life history characteristics of WA dhufish, with relevant biological parameters used for stock assessments summarised in Table 4.3.

Table 4.3. Summary of biological parameters for WA dhufish. F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters (von Bertalanffy)*		$L_t = L_\infty (1 - \exp(-k(t - t_0)))$ Estimated using age- and length-based catch curve model in L3Assess (Hesp 2023)
L_∞ (mm TL)	983 (F), 1119 (M)	Updated 2024
k (y^{-1})	0.12 (F), 0.11 (M)	Updated 2024
t_0 (years)	0	Not estimated
Maximum age (years)	41	Hesp et al. 2002a
Maximum size (mm TL)	1,220	Hesp et al. 2002a
Natural mortality, M (y^{-1})	0.11	Hesp et al. 2002a
Length-weight parameters		W (g) = a TL (mm) ^b
a	1.97×10^{-5}	Updated 2023
b	2.980	Updated 2023
Reproduction	Gonochoristic, multiple (batch) spawner, indeterminate fecundity	Hesp et al. 2002a
Maturity parameters		Logistic
A_{50} (years)	3.83 (F), 3.37 (M)	Fisher et al. 2010
A_{95} (years)	7.01 (F), 5.22 (M)	Fisher et al. 2010
L_{50} (mm)	331 (F), 324 (M)	Fisher et al. 2010
L_{95} (mm)	509 (F), 454 (M)	Fisher et al. 2010
Maximum batch fecundity	Up to 500,000 eggs per batch	Lenanton et al. 2009
Fecundity parameters (batch)		$BF = (a \text{ TL} - b)^3$
a	0.0841	Wise et al. 2007, Lenanton et al. 2009
b	10.432	Wise et al. 2007, Lenanton et al. 2009
Spawning duration	Up to 6 months per year (length-dependent)	Evans-Powell et al. 2024
Spawning frequency	Daily	Evans-Powell et al. 2024

* Note that the recent assessment has accounted for temporal variation in growth of WA dhufish by fitting a growth model that allows k to differ among years to available length at age data.

4.2.3.1 Habitats and Movements

WA dhufish inhabit waters up to 200 m deep and are closely associated with reef environments throughout their range (Hesp and Potter 2000). Small juveniles (<150 mm TL) have been found over sand-inundated reef margins adjacent to reefs, where there is an abundance of sponges or small patches of seagrass (Hesp et al. 2002a; Lewis et al. 2012). They have also been recorded around artificial habitats used for abalone sea ranching in waters off Augusta (Lewis et al. 2012). Larger juveniles (150-300 mm TL) are typically found over low-lying reefs, on rock ledges less than 30 cm high (Hesp et al. 2002a; Lewis et al. 2012).

At around the time they reach maturity, WA dhufish typically move to reef structures that are several metres in vertical height, including coral reefs at the Abrolhos Islands and limestone reefs in the temperate waters of the lower west coast (Hesp and Potter 2000; Hesp et al. 2002a; Lenanton et al. 2009). Tagging studies indicate that adult WA dhufish are largely sedentary and typically undergo only limited movements (Mackie et al. 2009). However, some individuals have been found to move tens of kilometres, primarily in an offshore-onshore pattern (Mackie et al. 2009).

Anecdotal reports suggest WA dhufish migrate into waters off Cape Naturaliste for spawning during summer, possibly as a strategy to optimise egg and larval retention by the Capes Current (Pearce and Pattiaratchi 1999; Strzelecki et al. 2013). While this area has been noted as a key spawning area in the past, aggregating individuals have been found across the WCB and are not limited to the spawning season (Mackie et al. 2009). Higher densities of WA dhufish reported in shallow coastal waters during late autumn and early spring may be associated with feeding (Lek 2004; Mackie et al. 2009).

4.2.3.2 Age and Growth

The method used for ageing WA dhufish, counting opaque zones in thin transverse sections of sagittal otoliths, has been validated using marginal increment analysis (Hesp et al. 2002a). Results showed that the mean monthly marginal increments increase from a low level in January to a maximum in September, before opaque zones are delineated from the otolith edge around October each year (Hesp et al. 2002a).

The maximum recorded age and length of WA dhufish is 41 years and 1,220 mm TL, respectively, with males attaining a greater maximum size than females (Hesp et al. 2002a). There is substantial individual variation in the lengths of WA dhufish of each sex at different ages, with 15-year-old females ranging in size from around 600 mm TL to over 900 mm TL.

Recent analyses of biological data for WA dhufish have indicated substantial temporal variation in the mean length at age of this species, with similar patterns observed across its distribution (Fairclough et al. 2021). The lengths at age of WA dhufish have been estimated using a traditional von Bertalanffy growth model for early years (1996-98), and a 'year-effects' growth model (similar to that described by Cottingham et al. 2016) for more recent years during which biological sampling has been undertaken close to annually (2002-21). These temporal variations in growth are accounted for in the stock assessment by allowing the von Bertalanffy growth coefficient (k) to differ among years (Fairclough et al. 2021).

4.2.3.3 Natural Mortality

The instantaneous rate of M of 0.11 y^{-1} currently used in assessments of WA dhufish was calculated by Hesp et al. (2002a) using the empirical equation of Hoenig (1983) for fish, derived from the relationship between M and maximum age for 84 fish stocks. Similar values of M for this species, based on the maximum recorded age of 41 years, have been obtained using several other age-related related empirical estimation methods (Wise et al. 2007; Hewitt and Hoenig 2005; Dureuil and Froese 2021).

4.2.3.4 Reproduction

WA dhufish is a functional gonochorist (i.e. separate sexes). Individuals of both sexes first attain maturity around 3-4 years of age, at approximately 300 mm TL (Hesp et al. 2002a; Wise et al. 2007; Lenanton et al. 2009; see Table 4.3). Prior to 2023, a MLL of 500 mm had been in place for this species, aimed at ensuring fish could spawn at least once before legally being caught and retained by fishers. That size limit resulted in large proportions of the catch being released, many of which would have died, particularly when caught in deeper waters (Hesp and Potter 2000; StJohn and Syers, 2005; Department of Fisheries 2016; Ryan et al. 2021). Hence the MLL for WA dhufish was removed in 2023.

WA dhufish have been found to form small social hierarchies during the spawning season. Spawning occurs in pairs or small groups of individuals, with larger males likely competing to fertilise eggs by controlling access to females (Mackie et al. 2009). Male dhufish have relatively small testes and produce lower volumes of sperm compared to polygamous species like Snapper that spawn in large aggregations (Hesp and Potter 2000; Cleary and Jenkins 2003). Although biological samples of WA dhufish have indicated that monthly sex ratios in each management area are slightly skewed towards males, video data have indicated that female biases can occur among social groups during the key spawning period (Mackie et al. 2009).

Female WA dhufish spawn multiple times in a single season and exhibit indeterminate fecundity (Hesp and Potter 2000; Hesp et al. 2002a). The eggs and larvae are initially pelagic, with settlement to a benthic habitat occurring once their eyes have adapted to low light, which starts to occur between 1 and 3 weeks post-hatch (Pironet and Neira 1998; Shand et al. 2001; Cleary and Jenkins 2003). Metamorphosis is completed after 40-45 days, at a larval length of approximately 8 mm (Pironet and Neira 1998).

While spawning in this species can extend from October to May, most activity occurs between December and March (Hesp and Potter 2000; Hesp et al. 2002a). Analyses of trends in mean monthly gonadosomatic indices, and the presence in catches of females with pre-spawning or spawning ovaries, have shown that spawning duration increases with female body size, with large mature females spawning far longer than small mature females (i.e. ~6 vs 2 months, respectively) (Hesp 1997; Evans-Powell et al. 2024). A birth date of 1 February, corresponding to the approximate midpoint of the spawning season, has been assigned to this species.

The relationship for WA dhufish in the WCB between batch fecundity and size of females is described in Table 4.3, with batches of ~5,900 to 534,000 eggs estimated for individuals ranging from 361 to 972 mm TL (Lenanton et al. 2009). On average, female WA dhufish are estimated to spawn close to daily during the spawning period (Evans-

Powell et al. 2024). Based on the size-related differences in spawning season duration and spawning frequency, a hyperallometric relationship exists between potential annual fecundity and female spawning biomass. In other words, large mature females are shown to contribute disproportionately more to population reproductive potential than small mature females (Evans-Powell et al. 2024). In addition, for a given fish length, average diameter of tertiary yolk granule oocytes increases with increasing latitude, indicating the potential for greater reproductive success further south (Evans-Powell et al. 2024).

4.2.3.5 Factors Affecting Recruitment and Other Biological Parameters

The recruitment of juvenile WA dhufish varies spatially and temporally, influenced by the spawning output of adults as well as the oceanographic conditions affecting the planktonic stage of eggs and larvae. Large cohorts of WA dhufish can play a significant role in the fishery for years, with four years of strong recruitment between 1993 and 1997 dominating catches of this species in the early 2000s (Wise et al. 2007). Inter-annual variability in recruitment to this stock is more pronounced towards the southerly latitudes of its range (Fairclough et al. 2021). While the factors affecting recruitment success in this species are not well understood, the strength of the Capes Current during the summer spawning months may impact the retention of larvae to the South-West management area in particular (Pearce and Pattiaratchi 1999; Lenanton et al. 2009).

Demersal species like WA dhufish has been assessed as having a medium sensitivity to the impacts of climate change (Caputi et al. 2015; Newman et al. 2024). Increasing water temperatures and changes in ocean currents have the potential to influence the distribution and biology of these species, including spawning and recruitment. Cheung et al. (2012) predicted a 110 km southward shift in the centroid of the distribution of WA dhufish along the west Australian coast by 2055. Anecdotal reports of increased catches of this species on the south coast of WA in recent years may be related to such effects and/or to the dispersal of early life history stages by unseasonal currents during events such as the marine heatwave of 2010/11.

4.2.3.6 Diet and Predators

Analyses of gut contents of WA dhufish collected from waters of the Mid-West and Metropolitan management areas have demonstrated that this species feeds mostly on teleosts, crustaceans and cephalopods (Lek 2004; Platell et al. 2010). WA dhufish lack any prominent teeth and consume their prey by suction feeding, engulfing their prey whole (Platell et al. 2010). There is a pronounced size-related change in the diet of this species, from mainly small crustaceans in juveniles to teleosts in larger fish (Lek 2004; Platell et al. 2010; Lewis et al. 2012). Likely predators of WA dhufish include large teleosts, sharks and marine mammals. The trophic level of WA dhufish is 4.3 (Froese and Pauly 2023).

4.2.3.7 Parasites and Diseases

Gonads of WA dhufish have often been found infected by a large nematode parasite of the family Philometridae (*Philometra* sp.), which represents the first record of this species from Australian waters (Hesp et al. 2002b). The prevalence of infection was greater among female than male dhufish, presumably related to differences in gonad

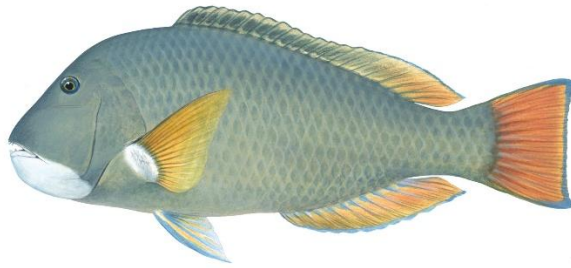
size. Reproductive analyses for WA dhufish, including histological examinations of gonads of fish infected by this parasite demonstrated that infection did not prevent fish from maturing, nor did it cause severe atrophy of the gonads or destruction of mature oocytes (Hesp et al. 2002b).

4.2.4 Inherent Vulnerability

The biological characteristics of WA dhufish make this species moderately vulnerable to fishing in the absence of appropriate management. Although individuals of this species mature around 3 years of age, they can attain a maximum age of at least 41 years. While the potential annual fecundity of WA dhufish is high, larger females produce disproportionately more eggs per spawning season than smaller mature females (Hesp 1997; Evans-Powell et al. 2024). As a species that exhibits highly variable recruitment, particularly at the southern end of its range, the presence of older age classes is important to ensure resilience to fishing pressure and sustain reproductive output.

The most common method of capture of demersal scalefish such as WA dhufish is by line fishing around reef habitats, with the majority of the catch landed by the recreational sector. Given their limited movements as adults, advances in fishing technology is likely to have led to increased susceptibility of this species to capture by fishers. Most WA dhufish are caught at depths >20 m, where they are highly susceptible to the effects of barotrauma (StJohn and Syers 2005). Research has shown that while the severity of barotrauma in this species increases with depth of capture, internal haemorrhaging can also be found in individuals irrespective of the depth in which they are caught (Ashby 1996; StJohn and Syers 2005). With large and thick-skinned swim bladders and blood oxygen properties characteristic of a species adapted to a relatively sedentary life at depth (Rogers et al. 1986; Stephens et al. 2002), WA dhufish do not cope well with the rapid changes in pressure experienced when pulled to the surface during capture. On average, 50% of released WA dhufish are expected to be susceptible to post-release mortality (DPIRD 2021). To reduce the need to release fish and minimise the impact of this additional source of mortality on stock recovery (Crisafulli et al. 2025), the MLL for WA dhufish was removed in 2023.

4.3 Baldchin Groper



4.3.1 Taxonomy and Distribution

Baldchin groper (*Choerodon rubescens*) is a member of the family Labridae (wrasses). Individuals range in colour from yellowish-brown as juveniles, to pinkish- or reddish-brown and greenish grey as adults, and are easily distinguished by the white chin and white pectoral peduncle (Hutchins and Swainston 1999). As a tuskfish, they have distinctive peg-like teeth protruding from both jaws (Lek et al. 2018).

Baldchin groper are endemic to WA and range from around Coral Bay in the north to Geographe Bay in the south (Hutchins and Swainston 1999). They are most abundant at the centre of their range, around the Houtman Abrolhos Islands (Hutchins 2001; Cure et al. 2018a).

4.3.2 Stock Structure

Baldchin groper comprise a single and genetically homogenous stock across the majority of its distribution (Fairclough et al. 2011; Gardner et al. 2015; Cure et al. 2018b). High levels of gene flow across its latitudinal range, and between offshore islands and the mainland, are facilitated by dispersal and mixing of pelagic eggs and larvae by the southward-flowing Leeuwin Current and northward-flowing counter currents (Cure et al. 2018b). Analyses of trace elements within the otoliths of Baldchin groper suggest that individuals recruit to the area where they reside as adults from around 1 year of age (Fairclough et al. 2011).

Recent genomic analyses have identified patterns of population structure of Baldchin groper consistent with local adaptation, where selective forces are likely to be impacting individuals at the cooler and warmer fringes of its distribution (Cure et al. 2018b). A strong recruitment to the southern parts of its range after the 2011 marine heatwave was found to have originated from spawning grounds several hundred kilometres to the north, suggesting a distributional shift to the south could be facilitated by warming waters and stronger Leeuwin Current flows (Cure et al. 2018b). As these individuals being transported southwards would be genetically adapted to temperatures that are higher than the waters they now inhabit, their ability to mature, spawn and survive at these southern latitudes remains uncertain (Cure et al. 2018c).

4.3.3 Life History

The sub-sections below provide an overview of the life history characteristics of Baldchin groper, with relevant biological parameters used for assessments summarised in Table 4.4.

Table 4.4. Summary of biological parameters for Baldchin groper in the North (Kalbarri/Mid-West areas) of the WCB. F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters (von Bertalanffy)		$L_t = L_\infty (1 - \exp(-k (t - t_0)))$ Estimated using age- and length-based catch curve model in L3Assess (Hesp 2023)
L_∞ (mm TL)	566 (sexes combined)	Updated 2024
k (y^{-1})	0.14 (sexes combined)	Updated 2024
t_0 (years)	0	Not estimated
Maximum age (years)	25 (28 in WA)	DPIRD biological data
Maximum size (mm TL)	745	DPIRD biological data
Natural mortality, M (y^{-1})	0.21	Wise et al. 2007
Length-weight parameters		W (g) = a TL (mm) ^b
a	2.25×10^{-5}	Updated 2023
b	2.977	Updated 2023
Reproduction	Protogynous hermaphrodite (female to male sex change), multiple (batch) spawner, indeterminate fecundity	
Maturity parameters		
A_{50} (years)	4.1 (F)	Fairclough 2005 (Abrolhos)
A_{95} (years)	6.2 (F)	Fairclough 2005 (Abrolhos)
L_{50} (mm)	279 (F)	Fairclough 2005 (Abrolhos)
L_{95} (mm)	352 (F)	Fairclough 2005 (Abrolhos)
Sex change parameters		
A_{50} (years)	11.9	Fairclough 2005 (Abrolhos)
A_{95} (years)	14.8	Fairclough 2005 (Abrolhos)
L_{50} (mm)	479	Fairclough 2005 (Abrolhos)
L_{95} (mm)	595	Fairclough 2005 (Abrolhos)
Fecundity	Up to 128,000 eggs per batch	Fairclough 2005 (Abrolhos)
Fecundity parameters (batch)		$BF = \exp(a \ln(TL) + b)$
a	5.05	Fairclough 2005 (Abrolhos)
b	-19.614	Fairclough 2005 (Abrolhos)

4.3.3.1 Habitats and Movements

Baldchin groper typically occur on, or in the vicinity of, benthic reef habitats up to 100 m deep (Fairclough et al. 2008; 2011). Juveniles are commonly found in mixed marginal habitats dominated by brown algae, seagrass, sand and rubble (Fairclough et al. 2008; 2011; Cure et al. 2015). These are thought to represent important settlement grounds for this species before moving into deeper coral or rocky reef environments (Cure et al.

2015). Movements of Baldchin groper are likely to occur at small spatial scales, with individuals remaining relatively resident to an area from around 1 year of age (Fairclough et al. 2011). Although aggregating behaviours of this species have been observed at the Abrolhos Islands during the spawning period, there was no evidence of spawning occurring at that time (Nardi 1999).

4.3.3.2 Age and Growth

Age estimation of Baldchin groper from thin transverse sections of sagittal otoliths has been validated using marginal increment analysis (Fairclough 2005), as well as by oxytetracycline hydrochloride marking of the otoliths (Nardi et al. 2006). Opaque zone delineation occurs in the spring and summer months, with the first zone typically becoming delineated by March of the second year of life at an age of ~15 months (Fairclough 2005).

Baldchin groper are relatively slow-growing, with the majority of growth occurring during the first 8 years of life and substantial variability around the lengths at all ages (Nardi et al. 2006). Although the mean TL of males at each age was typically greater than of females, there are no sex-specific differences in growth of this species.

4.3.3.3 Natural Mortality

The instantaneous rate of M currently used in the assessment of Baldchin groper, calculated by Wise et al. (2007) using the age-based empirical regression equation for fish (Hoenig 1983) and a maximum age of 22 years, is 0.21 y^{-1} . While some older individuals have since been recorded, the majority of these have been sampled from the southern part of the WCB.

4.3.3.4 Reproduction

Baldchin groper is a monandric protogynous hermaphrodite, with all fish first maturing as females before some later changing sex to males (Fairclough 2005; Nardi et al. 2006; Fairclough et al. 2023). The age at which individuals mature and change sex varies across its range, associated with latitudinal differences in growth patterns (Fairclough 2005). At the Abrolhos Islands, a female-biased sex ratio, sexual dichromatism, larger size of males, and small size of male testes relative to the ovaries of females, suggests that Baldchin groper has a polygynous mating system, where individuals form social groups that may be harem or lek-like, and where males are likely to fertilise the eggs of one female at a time (Fairclough 2005; Nardi et al. 2006; Fairclough et al. 2023).

Spawning of Baldchin groper occurs throughout its range, including at the Abrolhos Islands. Buoyant eggs released by a female are externally fertilised by the male, with the eggs and larvae dispersed by ocean currents for ~23 days before settling into suitable reef habitats as juveniles (Fairclough 2005; Cure et al. 2018c). Its spawning period at the Abrolhos Islands extends from July to January (Fairclough et al. 2023; Nardi et al. 2006), with most spawning activity occurring between October and December. The assigned birth date for this species, corresponding to the approximate midpoint of the spawning season, is 1 December (Fairclough 2005). Female Baldchin groper spawn on multiple occasions during a spawning season and, based on observations of hydrated oocytes in their ovaries, spawning occurs throughout the depth range from which sampling occurred. The relationship between batch fecundity and

length has been estimated for Baldchin groper at the Abrolhos Islands by Fairclough (2005; Table 4.4).

4.3.3.5 Factors Affecting Recruitment and Other Biological Parameters

Trends in age structure data indicate some variability in the recruitment of Baldchin groper between years. The survival of 0+ individuals is likely enhanced by the spring spawning of this species, leading to a long growing season. Increased water temperatures and ocean acidification have the potential to influence aspects of the biology of Baldchin groper, such as their spawning success and composition of their diet (Caputi et al. 2015). The 2010/11 marine heatwave resulted in observed mortalities of this species in shallow waters at the Abrolhos Islands (Pearce et al. 2011b). Research in the years following this heatwave event showed an unusually high recruitment towards the cooler edge of its range, where temperature anomalies were lowest, coupled with low recruitment at the warmer range edge (Cure et al. 2018a). Cheung et al. (2012) predicted that warming sea temperatures will shift the centroid of the distribution of Baldchin groper southwards by ~78 km by 2055, suggesting that the species will become less productive at the northern extent of its range and more so towards the southern extent. Baldchin groper has been assessed as having medium sensitivity to climate change effects (Newman et al. 2024).

4.3.3.6 Diet and Predators

Gut content analysis has demonstrated that the diet of Baldchin groper from the Abrolhos Islands consists predominantly of invertebrates (Lek et al. 2018). A change in diet was observed with increasing length of fish, from crustaceans, bivalves and gastropods in smaller juveniles to echinoderms in larger adults. Juveniles have been observed to feed by pectoral fin flapping to displace sand and find invertebrate food in shallow waters (Lek 2004; Cure et al. 2018c). Predators of Baldchin groper likely include larger teleosts, sharks and marine mammals. The trophic level of Baldchin groper is 3.6 (Froese and Pauly 2023).

4.3.3.7 Parasites and Diseases

While no published data exist, the gonads of Baldchin groper have frequently been found infected with nematode parasites and abdominal parasites (Fairclough unpublished data).

4.3.4 Inherent Vulnerability

The biological characteristics of Baldchin groper, including its moderate longevity and sex change, can make them vulnerable to the effects of fishing. With fishing typically being selective for larger individuals, it may disproportionately affect the relative abundance of male Baldchin groper in the population. This has the potential to influence egg fertilisation rates due to sperm limitation, given the small size of male testes and social spawning behaviour during the spawning season of this species. Baldchin groper are primarily caught by line fishing close to reef habitats, with the majority of catches landed by the recreational fishing sector. On average, 90% of Baldchin groper are expected to die following capture and release (DPIRD 2021).

4.4 Redthroat Emperor



4.4.1 Taxonomy and Distribution

Redthroat emperor (*Lethrinus miniatus*) is a member of the family Lethrinidae (emperors). Individuals of this species are readily distinguished by their red fins, throat and corner of mouth, with red-orange colouration around the eyes.

Redthroat emperor are widely distributed across the Indo-Pacific region, from southern Japan to northern Australia and New Caledonia (Carpenter and Niem 2001). The distribution of this species in Australia appears to be discontinuous, ranging from south of Perth to the Montebello Islands on the west coast, and between approximately 18° S and 24.5° S on the east coast (Williams 2003).

4.4.2 Stock Structure

Analysis of mitochondrial DNA has indicated there are two separate biological stocks of Redthroat emperor in western and eastern Australia (van Herwerden et al. 2009).

4.4.3 Life History

The sub-sections below provide an overview of the life history characteristics of Redthroat emperor, with estimated biological parameters used in assessments of the WA stock summarised in Table 4.5.

4.4.3.1 Habitats and Movements

Redthroat emperor generally occur on and around coral and rocky reefs (Carpenter 2001) and can be found at depths greater than 100 m (Newman and Williams 1996). The juvenile habitat of this species is relatively unknown. In WA, small individuals have occasionally been observed in marginal rubble habitats adjacent to exposed reefs in depths of around 20 m. Tagging studies on the east coast of Australia have yielded limited information about the movement patterns of Redthroat emperor (Williams et al. 2010). While the majority of fish recaptured within one year had remained on the same reef, most long-term recaptures were caught on different reefs, including a few individuals that moved distances of more than 100 km (Sawynok unpublished data; Williams et al. 2010). Analyses of $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ stable isotopes in otoliths indicate that this species exhibits variable ontogenetic movements along the Great Barrier Reef (Currey et al. 2014).

Table 4.5. Summary of biological parameters for Redthroat emperor in the North (Kalbarri/Mid-West areas) of the WCB. F = females.

Parameter	Value(s)	Comments/Source
Growth parameters (von Bertalanffy)		$L_t = L_\infty (1 - \exp(-k(t - t_0)))$ Estimated using age- and length-based catch curve model in L3Assess (Hesp 2023)
L_∞ (mm TL)	479 (sexes combined)	Updated 2024
k (y^{-1})	0.369 (sexes combined)	Updated 2024
t_0 (years)	0	Not estimated
Maximum age (years)	23 (27 in WA)	DPIRD biological data
Maximum size (mm TL)	679	DPIRD biological data
Natural mortality, M (y^{-1})	0.18	Hewitt and Hoenig 2005, based on max age of 23 years
Length-weight parameters		W (g) = a TL (mm) ^{<i>b</i>}
a	9.51×10^{-6}	Updated 2023
b	3.081	Updated 2023
Reproduction	Protogynous hermaphrodite (female to male sex change), multiple spawner	
Maturity parameters		
L_{50} (mm)	200 (F)	DPIRD unpublished
L_{95} (mm)	499 (F)	DPIRD unpublished
Sex change parameters		
L_{50} (mm)	465	DPIRD unpublished
L_{95} (mm)	585	DPIRD unpublished
Fecundity	No available data	

4.4.3.2 Age and Growth

Redthroat emperor is a medium-sized reef fish, with a maximum recorded size of 679 mm TL in WA. Biological studies using counts of validated annual increments in otoliths have revealed a maximum age of at least 20 years on the east coast of Australia (Brown and Sumpton 1998; Williams 2003; Williams et al. 2003), with the oldest aged individual in WA being 27 years. Redthroat emperor approach their maximum length at around 6 years of age, however, there is substantial variability in the length at age among individuals (Williams 2003) and locations (Brown and Sumpton 1998; Williams et al. 2003). Estimated parameters for von Bertalanffy growth curves and relationships between weight and total length of Redthroat emperor in WA, using available data of both sexes combined from the Kalbarri and Mid-West areas, are provided in Table 4.5.

4.4.3.3 Natural Mortality

An instantaneous rate of M of $0.18 y^{-1}$ has been calculated for Redthroat emperor in the northern WCB, using the mortality equation described by Hewitt and Hoenig (2005) and

a maximum recorded age within this region of 23 years (Table 4.5), noting that older individuals have been sampled in the southern WCB.

4.4.3.4 Reproduction

As a protogynous hermaphrodite, Redthroat emperor first mature as females and later in life change sex to males (Bean et al. 2003). Half of females attain maturity at around 200 mm TL in WA (DPIRD unpublished data; Table 4.5). While available data suggest that not all females of this species will change sex (Williams 2003; Sumpton and Brown 2004; DPIRD unpublished data), 50% are estimated to have changed to males at 465 mm TL in WA (Table 4.5).

A female-biased population on the east coast of Australia, which is typical for protogynous reef fish (Sadovy and Shapiro 1987), and the relatively small testes of males, suggests a polygynous mating pattern where a single male pair-spawns consecutively with several females. Available data from WA show that spawning in the northern parts of the WCB extends between October and February. With most spawning occurring between December and February, this species has an average birth date of 1 January in WA. Information on the early life history of this species is limited due to difficulties in identifying emperor larvae to the species level and a lack of information about the juvenile habitat.

4.4.3.5 Factors Affecting Recruitment and Other Biological Parameters

Due to the limited available information on the early life history of Redthroat emperor, potential factors affecting the recruitment of this species are not known. Age composition data for this species in WA indicates that recruitment varies substantially between years in the northern part of the WCB, with more consistent recruitment further north in the Gascoyne region (DPIRD unpublished data). In contrast to the east coast stock, the relatively low genetic diversity of Redthroat emperor in WA could suggest that this species is less resilient to future anthropogenic and natural disturbances, such as over-fishing and changing environmental conditions (van Herwerden et al. 2009). However, evidence of relatively strong recruitment around the time of the marine heatwave in 2010/11 and warm anomalies in 2012 and 2013 may indicate tolerance to temperatures that are above the long-term average (Lenanton et al. 2015). Cheung et al. (2012) predicted a 72 km southerly shift of the centroid of the Redthroat emperor range along the west coast of Australia by 2055. Redthroat emperor has been assessed as having medium sensitivity to climate change effects (Newman et al. 2024).

4.4.3.6 Diet and Predators

Redthroat emperor are generalist predators and consume a variety of crustaceans, echinoderms, molluscs and fish (Walker 1978). Predators of Redthroat emperor likely include large teleosts, sharks and marine mammals. The trophic level of Redthroat emperor is 4.2 (Froese and Pauly 2023).

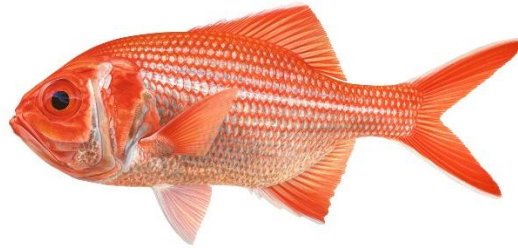
4.4.3.7 Parasites and Diseases

No published data exist on the occurrence of parasites in Redthroat emperor.

4.4.4 Inherent Vulnerability

Hermaphroditic species like Redthroat emperor can be inherently vulnerable to fishing, which typically selects for larger individuals and may disproportionately impact males. The moderate longevity of this species, and the variability in recruitment observed in the southern parts of its range in WA, suggests that fishing can be detrimental to the resilience of this stock in the absence of appropriate management. Redthroat emperor are primarily caught by commercial line fishers in the northern areas of the WCB, with a small proportion of the overall catches taken by recreational and charter fishers. Short-term survival of Redthroat emperor released into vertical net enclosures in experiments in Queensland, ranged from 76% to 92%, varying due to condition of the fish on release (Sumpton et al. 2008). While data on the proportion of individuals of this species in WA that are released following capture is limited, 25% of individuals are assumed to die post-release (DPIRD 2021).

4.5 Bight Redfish



4.5.1 Taxonomy and Distribution

Bight redfish (*Centroberyx gerrardi*) belongs to the family Berycidae (alfonsinos) and is also commonly referred to as red snapper or nannygai in WA. Individuals are silvery-red and characterised by a white lateral line, a red head with a white vertical band on the gill cover, a red eye, and white leading edges on the fins (Gomon et al. 2008). On the lower west coast of WA, the smaller-sized congeners *C. australis* (Yelloweye redfish) and *C. lineatus* (Swallowtail) are also caught but comprise only a small proportion of the overall 'redfish' catch composition (Mackie et al. 2009).

Bight redfish are endemic to southern Australia, where they are broadly distributed in deep waters along the edge of the continental shelf from Northern Tasmania in the east to Lancelin in the west (Hutchings and Swainston 1999). Within WA, this species is mostly caught on the south coast, however, they are also prominent in waters off Cape Naturaliste to Cape Leeuwin (the 'Capes' region) in the lower WCB.

4.5.2 Stock Structure

Examination of microsatellite DNA suggest Bight redfish likely comprises a single genetic stock across southern Australia (Bertram et al. 2015). Differences in otolith microchemistry signatures (stable isotopes and trace elements) between samples from the Great Australian Bight (GAB) and south-western WA, however, indicate limited mixing of adults between these regions (DPIRD unpublished data). To minimise the risk of localised depletion, assessments and management of Bight redfish in WA are currently undertaken at a bioregion level.

4.5.3 Life History

The sub-sections below provide an overview of the life history characteristics of Bight redfish, with estimated biological parameters used in assessments of this species in the WCB summarised in Table 4.6.

Table 4.6. Summary of biological parameters relevant to assessments of Bight redfish in the WCB.
F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters (von Bertalanffy)		$L_t = L_\infty (1 - \exp(-k(t - t_0)))$ Estimated using age- and length-based catch curve model in L3Assess (Hesp 2023)
L_∞ (mm TL)	575 (F), 559 (M)	Updated 2024
k (y^{-1})	0.1 (F), 0.12 (M)	Updated 2024
t_0 (years)	0 (assumed by method)	Not estimated
Maximum age (years)	84	DPIRD biological data (WA)
Maximum size (mm TL)	706	DPIRD biological data (WA)
Natural mortality, M (y^{-1})	0.05	Hewitt and Hoenig 2005, based on max age of 84 years
Length-weight parameters		W (g) = a TL (mm) ^{<i>b</i>}
a	7.73×10^{-6}	Mackie et al. 2009 (Capes, adjusted)
b	3.081	Mackie et al. 2009 (Capes, adjusted)
Reproduction	Gonochoristic, multiple (batch) spawner	
Maturity parameters		
A_{50} (years)	5-9 (F), 13.6 (M)	Coulson et al. 2018 (Capes)
A_{95} (years)	Not estimated (F), 46.9 (M)	Coulson et al. 2018 (Capes)
L_{50} (mm TL)	307 (F), 420 (M)	Coulson et al. 2018 (Capes)
L_{95} (mm TL)	696 (F), 697 (M)	Coulson et al. 2018 (Capes)
Fecundity	Up to 1.54 million eggs	Brown and Sivakumaran 2007 (GAB), based on standing stock of yolked, nuclear migrated and hydrated oocytes
Fecundity parameters (total)		Total fecundity = $a \times FL^b$
a	386.54	Brown and Sivakumaran 2007 (GAB)
b	2.16	Brown and Sivakumaran 2007 (GAB)

4.5.3.1 Habitats and Movements

Bight redfish are typically associated with reefs and can be found at depths of more than 200 metres (Gomon et al. 2008). Juveniles are commonly found in shallower, nearshore waters, while adults are most abundant along the edge of the continental shelf (Gomon et al. 2008). It is typically a solitary species (Hutchings and Swainston 1999) but occasionally form small schools, particularly during the spawning season when individuals have been observed aggregating over “lumps” on the seabed (Mackie et al. 2009). Anecdotal evidence from fishers suggests an inshore movement of Bight redfish coinciding with the start of the spawning season (Mackie et al. 2009).

4.5.3.2 Age and Growth

Bight redfish are very long-lived, with one individual caught off Albany in 2013 found to be 84 years old. The previously recorded maximum age of 71 years for this species was based on biological sampling in the GAB (Stokie and Krusic-Golub 2005). Opaque zones in sectioned otoliths of Bight redfish sampled in WA have been validated as deposited annually between June and January (Coulson et al. in prep.).

Estimated growth patterns of female and male Bight redfish in WA are similar (Mackie et al. 2009; Coulson et al. in prep.). Growth analyses based on extensive biological sampling across its range in WA between 2010 and 2015, suggest no marked differences in estimated growth parameters among areas (Coulson et al. in prep.).

4.5.3.3 Natural Mortality

Bight redfish have relatively low rates of population productivity (Wise and Tilzey 2000). Based on a maximum recorded age of 84 years in WA, the M estimated by the empirical methods of Hoenig (1983, fish equation) and Hewitt and Hoenig (2005; consistent with estimate from method by Dureuil and Froese 2021) is 0.05 y^{-1} .

4.5.3.4 Reproduction

The ages and lengths at which Bight redfish reach maturity vary markedly across its range in WA, with regional differences in reproductive potential possibly linked to variations in ocean productivity driven by upwelling during the spawning season (Coulson et al. 2019). In the Capes region of the WCB, an estimated 50% of females attain maturity at around 5-9 years of age (around 300 mm TL), compared to 12 years (189 mm TL) in the western South Coast Bioregion (SCB), and 21 years (512 mm TL) in the eastern SCB (Coulson et al. 2019).

Bight redfish are multiple batch spawners, with individuals aggregating to spawn in multi-male, multi-female groups (Mackie et al. 2009). Spawning of Bight redfish peaks when water temperatures are at their warmest in summer and autumn. In the WCB, spawning occurs from February through to at least April, with greatest spawning activity in March and April (Mackie et al. 2009; Coulson et al. 2019). A high proportion of spawning fish sampled during these months indicates that female Bight redfish spawn at frequent intervals whilst aggregated. Continuous distribution of whole oocyte diameters suggests that the fecundity of this species is indeterminate, with the number of oocytes to be spawned not being fixed prior to the onset of spawning (Brown and Sivakumaran 2007). Eggs and larvae are pelagic (Leis and Carson-Ewart 2000), however, the extent of dispersal is unknown.

4.5.3.5 Factors Affecting Recruitment and Other Biological Parameters

Regional differences in potential reproductive output of Bight redfish in WA are not consistent with the eastward decline in mean monthly water temperature. Recruitment may instead be related to summer upwelling in the Capes region, driving greater oceanic productivity prior to peak spawning that supports larval survival. In autumn, the prevailing southward- and eastward-flowing, downwelling Leeuwin Current strengthens, providing a dispersal mechanism along the west and south coasts. Bight redfish has a medium sensitivity to climate change effects in the WCB (Newman et al. 2024).

4.5.3.6 Diet and Predators

Bight redfish are carnivorous, with their diet in WA found to change with increasing body size, from crabs and isopods in smaller individuals to teleosts in adults (Platell et al. 2022). Likely predators of Bight redfish include large teleosts, sharks and marine mammals. The trophic level of Bight redfish is 4.1 (Froese and Pauly 2023).

4.5.3.7 Parasites and Diseases

No published data exist on the occurrence of parasites in Bight redfish.

4.5.4 Inherent Vulnerability

Long-lived species like Bight redfish have very low productivity and are inherently vulnerable to exploitation. Studies indicate that regional differences in biology and reproductive output of this species may be related to areas of upwelling. These characteristics support the precautionary approach taken to focus assessment and management in WA at finer spatial scales (i.e., at the bioregion level). Bight redfish are primarily caught by commercial line fishers on the south coast, with catches in the lower WCB taken primarily during the spawning season. Due to attaining maturity at an earlier age and smaller size in the Capes region, immature individuals (below the MLL of 300 mm TL) are rarely caught in the WCB. The post-release mortality of this species is estimated to be around 25%.

4.6 Breaksea Cod



4.6.1 Taxonomy and Distribution

Breaksea cod (*Epinephelides armatus*) is a member of the family Serranidae (groupers and rockcods). It varies in colour from yellow, pink or greenish to brown or grey, with a square tail and a distinctive black blotch surrounding its anus (Hutchins and Swainston 1999). This species is endemic to WA, ranging from around Shark Bay to the Recherche Archipelago (Hutchins and Swainston 1999).

4.6.2 Stock Structure

Studies of the stock structure of Breaksea cod have not been conducted. However, differences in the biological characteristics of this species between the south and west coasts of WA indicate limited mixing between these regions post-settlement (Buckland 2022).

4.6.3 Life History

The sub-sections below provide an overview of the life history characteristics of Breaksea cod, with relevant biological parameters used for stock assessments summarised in Table 4.7.

4.6.3.1 Habitats and Movements

Breaksea cod inhabit rocky reefs in waters up to 100 m deep (Gomon et al. 2008). Coastal reef habitats also provide important nurseries for this species. Juveniles <100 mm in total length exhibit cryptic behaviour by swimming upside down under cave roofs and ledges, likely providing refuge from predators (Fairclough 2016).

4.6.3.2 Age and Growth

Marginal increment analysis demonstrated that a single opaque zone is formed annually in the otoliths of Breaksea cod and that the numbers of these zones can be used for estimating the age of individuals (Moore et al. 2007). Consistent with the Metabolic Theory of Ecology (MTE; van der Meer 2006), growth analyses for this species indicate that individuals on the south coast of WA grow to a larger size than those on the west coast (Buckland 2022). Parameters describing the von Bertalanffy growth of females were re-estimated from recently collected data using an age- and length-based catch curve model that simultaneously estimates growth, fishing mortality and length-based selectivity (Table 4.7).

Table 4.7. Summary of biological parameters relevant to assessments of Breaksea cod in the WCB. F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters (von Bertalanffy)		$L_t = L_\infty (1 - \exp(-k (t - t_0)))$ Estimated using age- and length-based catch curve model in L3Assess (Hesp 2023)
L_∞ (mm TL)	457 (F)	Updated 2024
k (y^{-1})	0.3 (F)	Updated 2024
t_0 (years)	0	Not estimated
Maximum age (years)	22 (WCB)	Buckland 2022
Maximum size (mm TL)	565 (WCB)	Buckland 2022
Natural mortality, M (y^{-1})	0.19	Hoenig 1983; Hewitt and Hoenig 2005
Length-weight parameters		W (g) = a TL (mm) ^b
a	8.02×10^{-6}	Updated 2023
b	3.115	Updated 2023
Reproduction	Gonochoistic, Multiple (batch) spawner, indeterminate fecundity	
Maturity parameters		
L_{50} (mm TL)	306 (F), 256 (M)	Moore et al. 2007
L_{95} (mm TL)	357 (F), 317 (M)	Moore et al. 2007
Fecundity	No available data	

4.6.3.3 Natural Mortality

Based on a maximum age of 22 years for Breaksea cod on the west coast of WA (Buckland 2022), an estimated M of $0.19 y^{-1}$ for this species is derived using the methods described by Hoenig (1983) and Hewitt and Hoenig (2005), where the latter corresponds to approximately 1.5% of individuals surviving to the maximum age.

4.6.3.4 Reproduction

Unlike most serranids, Breaksea cod has been confirmed to be gonochoristic rather than hermaphroditic (Moore et al. 2007). Based on data collected from the WCB in the early 2000s, the majority of female and male Breaksea cod are estimated to have matured by 7 and 5 years of age, respectively (Moore et al. 2007). An estimated 50% of females and males attain maturity by 306 and 256 mm TL, respectively (Moore et al. 2007; Table 4.7).

Spawning of Breaksea cod occurs between October and April, with the capture of females with mature or spawning ovaries in all months except between July and September indicating that spawning activity in this species occurs over a very protracted

period (Moore et al. 2007). Breaksea cod is a multiple spawner, with females releasing batches of eggs multiple times during the spawning period (Moore et al. 2007). The much greater size of female ovaries compared to male testes, and the larger length of males at any given age compared to females, suggest that Breaksea cod is a pair spawner. The eggs and larvae of Breaksea cod are pelagic (Neira et al. 1998; Leis and Carson-Ewart 2000).

4.6.3.5 Factors Affecting Recruitment and Other Biological Parameters

As age data for Breaksea cod have only been collected periodically to date, the extent of interannual recruitment variation is not well understood.

4.6.3.6 Diet and Predators

The diet of Breaksea cod consists mainly of small teleosts and crustaceans (Platell et al 2010). This species apparently employs ram feeding, with its large gape and prominent backward-curving canines presumably representing adaptations for the capture and retention of fish prey (Platell et al. 2010). The trophic level of Breaksea cod is 3.7 (Froese and Pauly 2023).

4.6.3.7 Parasites and Diseases

No published data exist on the occurrence of parasites in Breaksea cod.

4.6.4 Inherent Vulnerability

The moderate longevity and relatively late maturity of Breaksea cod make them vulnerable to the effects of fishing. Given the similarity of the estimated length at 50% female maturity with the MLL of 300 mm TL in place for this species until 2023, around half of females would have been able to be retained before attaining maturity. The larger size attained by males than females would also result in a greater susceptibility of males to size-selective fishing.

Breaksea cod are primarily caught by recreational line fishers in the WCB, with only small catches taken by commercial fishers. As individuals of this species possess a large mouth gape relative to their size, and as juveniles and adults are found in the same habitat, substantial numbers of small fish tend to be caught by anglers (Crisafulli et al. 2024). These two sectors operate across the majority of the known distribution of this species, using methods to specifically target demersal fish. The MLL for this species was removed in 2023 to reduce release rates of Breaksea cod, with around 50% of fish expected to be susceptible to mortality following release (DPIRD 2021).

4.7 Hapuku



4.7.1 Taxonomy and Distribution

Hapuku (*Polyprion oxygeneios*) is a member of the family Polyprionidae (wreckfishes), which also includes Bass groper (*Polyprion americanus*) (see Section 4.8). These only two species of the genera *Polyprion* appear to be sympatric throughout their distribution in the southern hemisphere (Williams et al. 1996; Barreiros et al. 2004). Hapuku varies in colour from silvery blue to pinkish brown, typically with a darker upper part of its body (Hutchins and Swainston 1999). It can be distinguished from Bass groper by its more slender body, concave caudal fin, smaller eye and more projecting lower jaw (Barreiros et al. 2004). In Australia, Hapuku are known to occur off the southern coast from around Rottnest Island in WA to Sydney in the east, and around Tasmania (Hutchins and Swainston 1999).

4.7.2 Stock Structure

Results from recent analyses of mitochondrial and microsatellite DNA of Hapuku sampled across the southern hemisphere has led to the rejection of an earlier hypothesis of overall pan-oceanic mixing of this species (Stewart 2020). While the study demonstrated genetic differentiation between fish from different ocean basins, Hapuku was found to be genetically homogenous across Australia (Stewart 2020).

4.7.3 Life History

The sub-sections below provide an overview of the life history characteristics of Hapuku in WA, with biological parameters used for stock assessments summarised in Table 4.8.

4.7.3.1 Habitats and Movements

Hapuku has an unusual life history strategy that involves an extended pelagic juvenile stage of up to 4 years in oceanic waters (Roberts 1989; 1996). Prior to their settlement, pelagic juvenile Hapuku have been observed in clear oceanic waters near floating objects such as logs, buoys and rafts of algae (Roberts 1996). It is unclear if juveniles actively swim in these surface waters or passively drift when associated with floating objects. Sub-adult and adult Hapuku are found in demersal reef habitats, predominantly in mid-shelf to upper slope waters at depths between 50 m and 850 m (Barreiros et al. 2004; Wellington et al. 2021). Tagging studies have shown that some individuals of this species migrate large distances associated with spawning (Beentjes and Francis 1999).

Table 4.8. Summary of biological parameters relevant to assessments of Hapuku in southern WA. F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters		$L_t = L_\infty (1 - \exp(-k (t - t_0)))$
L_∞ (mm TL)	890 (sexes combined)	Wakefield et al. 2010
k (y^{-1})	0.24 (sexes combined)	Wakefield et al. 2010
t_0 (years)	-0.63 (sexes combined)	Wakefield et al. 2010
Maximum age (years)	52	Wakefield et al. 2010
Maximum size (mm TL)	1,114	Wakefield et al. 2010
Natural mortality, M (y^{-1})	0.08	Hoenig 1983; Hewitt and Hoenig 2005
Length-weight parameters		W (g) = a TL (mm) ^b
a	2.582×10^{-6}	Wakefield, unpublished data
b	3.24337	Wakefield, unpublished data
Reproduction	Gonochoristic, broadcast spawner	Wakefield et al. 2010
Maturity parameters		
A_{50} (years)	7.1 (F), 6.8 (M)	Wakefield et al. 2010
A_{95} (years)	9.5 (F), 10.1 (M)	Wakefield et al. 2010
L_{50} (mm TL)	760 (F), 702 (M)	Wakefield et al. 2010
L_{95} (mm TL)	851 (F), 774 (M)	Wakefield et al. 2010
Fecundity	No available data	

4.7.3.2 Age and Growth

Hapuku can grow to over 1,000 mm TL and reach a maximum age of at least 52 years in WA (Wakefield et al. 2010). A single opaque zone has been found to be deposited annually in the otoliths of Hapuku from the south coast of WA, beginning in June and completed in all fish by the end of October (Wakefield et al. 2010).

The growth rates of females and males vary throughout their life, with individuals initially growing quickly at ~150-300 mm TL per year during their pelagic life stage, before slowing markedly between the ages of 2 and 5 years as individuals settle in benthic habitats. The estimated growth rate of Hapuku subsequently increased in the years following this transition, around the time when individuals attain maturity, before slowing at around 10 years of age once all fish were mature (Wakefield et al. 2010). Due to the observed suppression of growth during settlement, von Bertalanffy growth curves for Hapuku on the southern coast of WA were only fitted to data for fish ≥ 5 years, with no biologically significant differences in growth parameters found between females and males (Wakefield et al. 2010; Table 4.8).

4.7.3.3 Natural Mortality

The value of M for Hapuku, based on a maximum recorded age of 52 years in southern WA and using two empirical estimation methods (Hoenig 1983; Hewitt and Hoenig 2005) has been estimated as 0.08 y^{-1} .

4.7.3.4 Reproduction

Hapuku is a gonochoristic, broadcast spawning species. On average, individuals of both sexes attain maturity around 7 years of age, at a slightly larger size for females than males (Wakefield et al. 2010; Table 4.8). Spawning off the south coast of WA extends from May to September (Wakefield et al. 2010). Mass movements of Hapuku recorded around the time of spawning may be indicative of this species forming spawning aggregations (Beentjes and Francis 1999).

4.7.3.5 Factors Affecting Recruitment and Other Biological Parameters

Potential factors affecting the recruitment of Hapuku are not known. Impacts of climate change on offshore demersal species like Hapuku are expected to be lower than inshore demersal species, however, increasing sea surface temperatures and shifts in ocean currents could influence this species during its pelagic juvenile phase. This species has been assessed as having medium sensitivity to climate change (Newman et al. 2024).

4.7.3.6 Diet and Predators

Hapuku are carnivorous and feed on a wide range of prey, including teleost fish and squid. While juveniles are likely to be vulnerable to predation by a wide range of fish, marine mammals and sea birds, adults have been found to be predated on by sperm whales, orcas and white sharks (Roberts 1996). The trophic level of Hapuku is 4.5 (Froese and Pauly 2023).

4.7.3.7 Parasites and Diseases

No published data exist on the occurrence of parasites in Hapuku.

4.7.4 Inherent Vulnerability

Long-lived and late-maturing species like Hapuku are inherently vulnerable to fishing pressure. Given the extended pelagic phase of juveniles, however, immature individuals are only susceptible to demersal line fishing for a relatively short period between settlement and maturity. Although demersal line fishing for offshore species like Hapuku occurs throughout their known distribution in WA, catches fluctuate widely between years as fishing effort focused on this deep-water offshore resource is limited by weather conditions particularly off the southern coastline. Available data suggests that discarded catches of Hapuku are negligible (Ryan et al. 2021).

4.8 Bass Groper



4.8.1 Taxonomy and Distribution

Bass groper (*P. americanus*) belongs to the family Polyprionidae (wreckfishes). It largely overlaps in its distribution across the southern hemisphere with the congeneric Hapuku (Williams et al. 1996; Barreiros et al. 2004; see Section 5.7). In southern Australia, Bass groper has been found in waters from around Fraser Island in Queensland to north of Perth in WA, and around Tasmania (Gomon et al. 2008). Bass groper varies in colour from dark grey to steely blue, with or without silvery blotches (Hutchins and Swainston 1999). It is deeper-bodied than Hapuku (Barreiros et al. 2004) and has a bony lump on the top of its head (Hutchins and Swainston 1999).

4.8.2 Stock Structure

A recent study of the population structure of Bass groper across the southern hemisphere, using both mitochondrial and microsatellite DNA markers, detected genetic differences between individuals sampled from the south-west of WA and those from eastern Australia and New Zealand (Stewart 2020).

4.8.3 Life History

The sub-sections below provide an overview of the life history characteristics of Bass groper in south-west WA, with biological parameters used for stock assessments summarised in Table 4.9.

4.8.3.1 Habitats and Movements

Like Hapuku, juvenile Bass groper are pelagic and typically associate with flotsam on the surface of oceanic waters. Following settlement to demersal habitats at around 5 years of age, adults are found in offshore waters around rocky reefs and structures in depths up to 1,000 m (Roberts 1996; Wellington et al. 2021). It is possible that Bass groper, like Hapuku, undertake seasonal migrations around the time of spawning (Roberts 1996).

Table 4.9. Summary of biological parameters relevant to assessments of Bass groper in south-western WA. F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters		$L_t = L_\infty (1 - \exp(-k (t - t_0)))$
L_∞ (mm TL)	1259 (F), 1113 (M)	Wakefield et al. 2013c
k (y^{-1})	0.09 (F), 0.10 (M)	Wakefield et al. 2013c
t_0 (years)	-0.36 (F), -1.54 (M)	Wakefield et al. 2013c
Maximum age (years)	78	Wakefield et al. 2013c
Maximum size (mm TL)	1,452	Wakefield et al. 2013c
Natural mortality, M (y^{-1})	0.05	Hoenig 1983; Hewitt and Hoenig 2005
Length-weight parameters		W (g) = a TL (mm) ^b
a	7.819×10^{-5}	Wakefield et al. 2013c
b	2.8028	Wakefield et al. 2013c
Reproduction	Gonochoristic, broadcast spawner	Wakefield et al. 2013c
Maturity parameters		
A_{50} (years)	14.4 (F), 11.2 (M)	Wakefield et al. 2013c
L_{50} (mm TL)	944 (F), 808 (M)	Wakefield et al. 2013c
L_{95} (mm TL)	1147 (F), 897 (M)	Wakefield et al. 2013c
Fecundity	No available data	

4.8.3.2 Age and Growth

Bass groper grow to a maximum length and age in south-west WA of at least 1,500 mm TL and 78 years, respectively (Wakefield et al. 2013c). Counts of opaque zones from transverse sections of the sagittal otoliths have been used in calculating the ages of Bass groper from south-western Australia, where a single opaque zone is deposited each year from May to September (Wakefield et al. 2013c). Estimated von Bertalanffy growth curves differ significantly between sexes, with females attaining larger lengths at age than males above 10 years (Wakefield et al. 2013c).

4.8.3.3 Natural Mortality

The estimated M of Bass groper in WA of $0.05 y^{-1}$ has been derived from Hoenig's (1983) fish equation and the equation by Hewitt and Hoenig (2005), based on the maximum recorded age of 78 years.

4.8.3.4 Reproduction

Bass groper in WA is a gonochorist, with 50% of females and males of this species estimated to have matured at 944 and 808 mm TL, respectively (Wakefield et al. 2013c; Table 4.9). Calculated using the inverse von Bertalanffy growth equation, these lengths

correspond to estimated ages at 50% maturity of around 14 years for females and 11 years for males (Wakefield et al. 2013c). Bass groper in south-west WA spawn between March and June, likely forming spawning aggregations (Wakefield et al. 2013c).

4.8.3.5 Factors Affecting Recruitment and Other Biological Parameters

Factors affecting the recruitment of Bass groper are unknown. While impacts of climate change on offshore demersal species is expected to be lower than inshore demersal species, increasing sea surface temperatures and shifts in ocean currents could influence this species during its pelagic juvenile phase. Bass groper is assessed as having medium sensitivity to climate change effects (Newman et al. 2024).

4.8.3.6 Diet and Predators

Bass groper are opportunistic and generalised carnivores that feed on a wide range of prey, with large adults having very few natural predators (Roberts 1996). The trophic level of Bass groper is 4.1 (Froese and Pauly 2023).

4.8.3.7 Parasites and Diseases

No published data exist on the occurrence of parasites in Bass groper.

4.8.4 Inherent Vulnerability

The exceptional longevity and late maturation of Bass groper makes it inherently vulnerable to fishing, with a low production potential and limited resilience to exploitation. The lower relative abundance of Bass groper in commercial and recreational catches from waters between 250-600 m deep compared to other offshore species sampled concurrently may suggest that this species is more common in deeper waters than those typically fished by these sectors. The extended pelagic phase of juvenile Bass groper largely protects immature individuals of this species from exploitation by demersal fisheries in south-west Australia. Available data suggests that discarded catches of Bass groper are negligible.

4.9 Blue-Eye Trevalla



4.9.1 Taxonomy and Distribution

Blue-eye trevalla (*Hyperoglyphe antarctica*) is a member of the family Centrolophidae (trevallas). It varies in colour from silvery blue to pinkish brown, often with a darker upper part of its body, and has a large deep-blue eye with a golden ring around it (Hutchins and Swainston 1999). Blue-eye trevalla are widely distributed in deep offshore waters of the southern hemisphere. In Australia, this species occurs from New South Wales in the east to around Perth in the west, and around Tasmania (Hutchins and Swainston 1999).

4.9.2 Stock Structure

Based on phenotypic variation in age and growth, otolith microchemistry and potential larval dispersal, four geographically distinct subpopulations of Blue-eye trevalla have been distinguished around the Australian coast, with the western stock comprising fish from WA to western Tasmania (Williams et al. 2017).

4.9.3 Life History

The sub-sections below provide an overview of the life history characteristics of Blue-eye trevalla, with biological parameters used for stock assessments in WA summarised in Table 4.10.

4.9.3.1 Habitats and Movements

Juvenile Blue-eye trevalla can be found in surface waters, with small individuals <100 mm often associating with drifting flotsam or algae (Duffy et al. 2000). In New Zealand, the transition to demersal habitats has been found to occur at around 2 years of age and 500 mm in length (Horn 1988). Once settled, Blue-eye trevalla associate with rocky ground on continental slopes, usually at depths of 200-900 m. Mature fish are thought to move into shallower waters and aggregate over specific areas for spawning (Baelde 1995).

4.9.3.2 Age and Growth

A single opaque zone has been found to form on the sagittae of Blue-eye trevalla each year (Horn 1988). The longevity of this species is poorly understood, with estimates ranging from around 30 to 76 years, where the latter was estimated using radiocarbon dating methods (Horn et al. 2010). A 32 year old individual was captured off Esperance in 2005, becoming the oldest individual recorded in WA waters (Wakefield and Newman 2008). Estimates of von Bertalanffy growth parameters for Blue-eye trevalla in western Tasmania, while based on data that lacked small individuals, are provided in Table 4.10.

Table 4.10. Summary of biological parameters relevant to assessments of Blue-eye trevalla in WA.
F = females, M = males.

Parameter	Value(s)	Comments/Source
Growth parameters		$L_t = L_\infty (1 - \exp(-k(t - t_0)))$
L_∞ (mm TL)	676 (combined sexes)	Williams et al. 2016 (Tasmania)
k (y^{-1})	0.16 (combined sexes)	Williams et al. 2016 (Tasmania)
t_0 (years)	-6.049 (combined sexes)	Williams et al. 2016 (Tasmania)
Maximum age (years)	32	Wakefield and Newman 2008 (WA)
Maximum size (mm TL)	1,066	Wakefield and Newman 2008 (WA)
Natural mortality, M (y^{-1})	0.13	Hoenig 1983, Hewitt and Hoenig 2005
Length-weight parameters		W (g) = a TL (mm) ^b
a	1.5897×10^{-5}	Wakefield unpublished data (WA)
b	2.9908	Wakefield unpublished data (WA)
Reproduction	Broadcast spawner	
Maturity parameters		
A_{50} (years)	11-12 (F), 8-9 (M)	Baelde 1995 (Tasmania)
L_{50} (mm TL)	713 (F), 616 (M)	Baelde 1996 (Tasmania)
Fecundity (annual)	Up to 11 million eggs per year	Baelde 1996 (Tasmania)
Fecundity parameters		F (millions) = a FL (cm) ^b
a	4.2×10^{-8}	Baelde 1996 (Tasmania)
b	4.22	Baelde 1996 (Tasmania)

4.9.3.3 Natural Mortality

Based on a maximum recorded age of 32 years in WA, an estimated M of $0.13 y^{-1}$ has been derived for Blue-eye trevalla using Hoenig's (1983) fish equation and the equation described by Hewitt and Hoenig (2005).

4.9.3.4 Reproduction

It is unclear if Blue-eye trevalla are gonochoristic or hermaphroditic, with males of the closely related *H. perciformis* found to contain both ovarian and testicular tissues (Filer and Sedberry 2008). The lengths at which 50% of female and male Blue-eye trevalla attain maturity have been estimated as 713 and 616 mm TL, respectively, which correspond to ages of around 11-12 years and 8-9 years for these two sexes (Baelde 1995; 1996).

Blue-eye trevalla spawn between November and May in southern WA (DPIRD unpublished data). The fecundity of Blue-eye trevalla has been found to be determinate, with females estimated to produce up to 11 million eggs annually, released in multiple

batches over the spawning season (Baelde 1995). The eggs, larvae and juveniles of this species are pelagic (Horn 1988).

4.9.3.5 Factors Affecting Recruitment and Other Biological Parameters

Factors affecting the recruitment of Blue-eye trevalla are unknown. While impacts of climate change on offshore demersal species is expected to be lower than inshore demersal species, increasing sea surface temperatures and shifts in ocean currents could influence this species during its pelagic juvenile phase. Blue-eye trevalla is assessed as having low sensitivity to climate change effects (Newman et al. 2024).

4.9.3.6 Diet and Predators

Diets of Blue-eye trevalla include cephalopods, crustaceans and fish (Horn and Massey 1989). Predators likely include sharks and marine mammals such as orcas. The trophic level of Blue-eye trevalla is 4.0 (Froese and Pauly 2023).

4.9.3.7 Parasites and Diseases

No published data exist on the occurrence of parasites in Blue-eye trevalla.

4.9.4 Inherent Vulnerability

The potentially high longevity Blue-eye trevalla and its late maturation makes it inherently vulnerable to exploitation. While its pelagic juvenile phase likely protects the species from demersal fishing in its first few years of life, it is likely that immature individuals are still regularly selected by line fishing in offshore waters of south-west WA. Available data suggests that discarded catches of Blue-eye trevalla are negligible.

5 Fishery Information

5.1 Fisheries Capturing the West Coast Demersal Scalefish Resource

The WCDSR is primarily targeted by commercial and recreational line-fishing sectors, where the latter comprises private and charter recreational fishers. The two sectors operate in the four main management areas of the WCB:

- Kalbarri (27°S¹ to 28°S),
- Mid-West (28°S to 31°S),
- Metropolitan (31°S to 33°S), and
- South-West (33°S to 115°30'E).

While the majority of fishing effort is focused on inshore waters of around 20-100 m depth, offshore demersal species are also targeted in deeper habitats.

The main commercial fishery targeting demersal species in the WCB is the West Coast Demersal Scalefish (Interim) Managed Fishery (WCDSIMF), which uses handlines and droplines to take around 90% of the total annual commercial catches. Demersal scalefish in the WCB are also caught as byproduct by the West Coast Demersal Gillnet and Demersal Longline (Interim) Managed Fishery and the Southern Demersal Gillnet and Demersal Longline Managed Fishery (collectively referred to as the Temperate Demersal Gillnet and Demersal Longline Fisheries, TDGDLF), which primarily target sharks using gillnets. The WCDSIMF and TDGDLF are prohibited from operating in the Metropolitan area since it was closed to commercial fishing for demersal scalefish in November 2007. Minor catches of demersal scalefish are also landed by several other commercial fisheries in the WCB, including the West Coast Rock Lobster Managed Fishery (WCRLMF) and the Cockburn Sound Line and Pot Managed Fishery (CSLPMF).

Recreational and charter fishers that target demersal species in the WCB almost exclusively fish from boats. While line fishing is the main method used by these fishers, some demersal species are also caught by spear fishing. In 2020-21, 78% and 22% of demersal scalefish catches collectively retained by the boat-based recreational sector in the WCB were landed by private recreational and charter vessels, respectively.

In addition to State-managed fisheries, the Commonwealth-managed Western Deepwater Trawl Fishery (WDTF) can operate in WA waters outside of the 200 m isobath and thus has the potential to interact primarily with offshore demersal species in the WCB. Catch and effort reporting for this fishery suggest that the level of effort in recent years has been low, with vessels primarily targeting bugs and Ruby snapper (Patterson et al. 2022).

The sections below provide more detailed information about the main fisheries that target the WCDSR, i.e. the WCDSIMF and the recreational and charter fisheries.

¹ Note that the Kalbarri management area of the WCDSIMF extends north to 26°30'S.

5.2 West Coast Demersal Scalefish (Interim) Managed Fishery

5.2.1 History of Development

Records of commercial fishing for demersal scalefish in WA date back to the late 1800s, with early fishing effort primarily focused out of Fremantle (Fraser 1898). The first targeted line fishery for Snapper developed in waters off Shark Bay in the early 1900s, when sailing vessels started travelling north from Geraldton each year to fish the winter spawning aggregations (Moran and Jenke 1989). By the 1930s, around 50 wetline vessels were operating in the waters around Geraldton and the Abrolhos Islands (Cooper 1996). The open access arrangements for this fishery in those early years make historical catches difficult to quantify (Crowe et al. 1999).

Compulsory logbook reporting of commercial catch and effort data since 1975 shows approximately decadal increases in handline and dropline catches of demersal scalefish in the late 1980s, 1990s and mid-2000s, peaking at more than 1,000 t in 2004 (Figure 5.1). More than 500 individual wetline vessels reported catches of demersal scalefish in the WCB between 1991 and 2003 (Department of Fisheries 2005), with the majority of catches landed over this period comprising Snapper from the Kalbarri and Mid-West areas. During this time, the fishery was regulated primarily through minimum size limits, gear limits and closed seasons for some species.

Following a review of open access wetline fishing on the west coast of WA (Department of Fisheries 2006), and the first WCDSR assessment confirming that overfishing of key stocks had been occurring (Wise et al. 2007), a suite of new management measures were implemented to reduce commercial catches of demersal scalefish in the WCB by at least 50% of the 2005-06 catches (i.e., below 450 t). The first of these measures was implemented in late 2007, when the Metropolitan area was closed to commercial fishing for demersal scalefish. The establishment of the WCDSMF on 1 January 2008 limited access to the WCDSR to 61 licenced commercial line fishing vessels, with maximum effort limits introduced for each of the management areas of the fishery the following year.

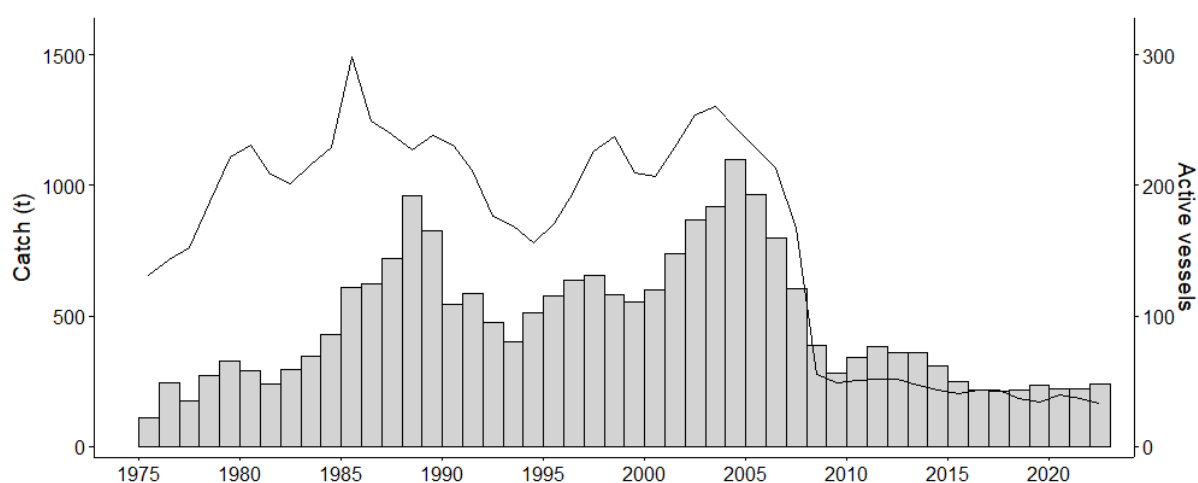


Figure 5.1. Annual retained commercial line fishing catches of demersal scalefish (t, grey bars) and number of active commercial line fishing vessels (black line) in the WCB since 1975.

Since the introduction of the WCDSIMF in 2008, commercial catches of demersal scalefish in the WCB have been maintained below the initial recovery benchmark of 450 t collectively set for the WCDSIMF and TDGDLF (Fairclough et al. 2021). After the significant reduction in catch resulting from the new management measures, an increase in fishing effort was observed between 2009 and 2012 (Figure 5.2), driven primarily by a strong recruitment of Snapper in the Mid-West and Kalbarri. Further reductions in effort entitlement to WCDSIMF fishers in these areas in 2015 contributed to the catches of Snapper being reduced below the species-specific recovery benchmark for this species.

The number of active vessels in the WCDSIMF has decreased from around 50 in 2008 to 30 in 2022, with the majority operating in the Mid-West area of the WCB (Figure 5.2). Since 2015, the number of days and hours fished have remained relatively stable, particularly in the Kalbarri and South-West management areas. A smaller increase in fishing activity was observed in the Mid-West during 2020, when impacts of the COVID-19 pandemic on the export markets of Western rock lobster resulted in a shift in effort onto demersal scalefish stocks. Many of the WCDSIMF fishers also hold licences in other commercial fisheries, with only a small proportion of the vessels operating as full-time wetline fishers.

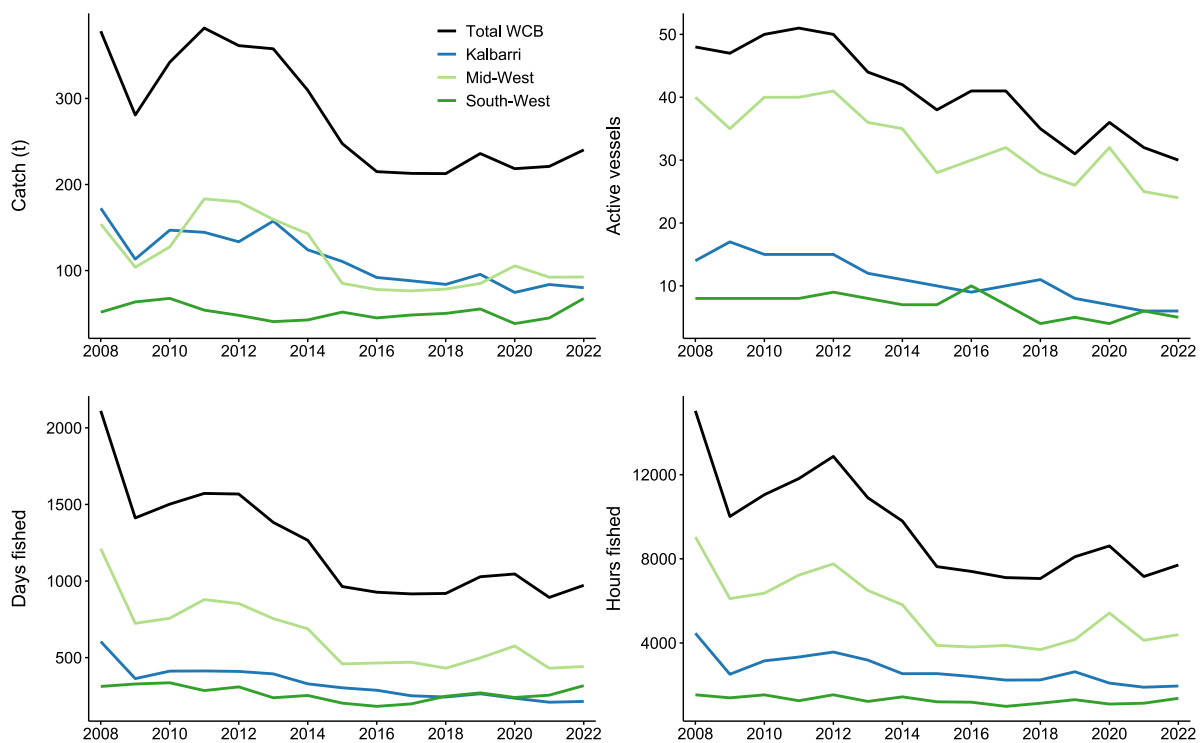


Figure 5.2. Annual demersal scalefish catch (t), number of active vessels, days fished and hours fished in the WCDSIMF since 2008.

5.2.2 Current Fishing Activities

A summary of key attributes of the WCDSIMF and its current fleet is provided in Table 5.1. The WCDSIMF encompasses the waters of the Indian Ocean from north of Kalbarri (26°30'S) to east of Augusta (115°30'E), extending seaward to the 200 nm boundary of the Australian Fishing Zone (Figure 5.3). Fishers in the WCDSIMF are not permitted to operate within the Metropolitan area, in the Reef Observation Areas of the Abrolhos FHPA, or in any sanctuary zones of State and Commonwealth Marine Parks within the boundaries of the fishery. WCDSIMF catches of demersal scalefish are primarily taken in depths of around 40-60 m along the edge of the continental shelf, with some fishing for deeper-water species also occurring in the South-West area.

Following an assessment in 2021 that demonstrated stocks of Snapper and WA dhufish were not recovering, a further 50% reduction in the WCDSIMF recovery benchmark to 218.5 t led to reduced entitlement (fishing hours) in all management areas. Additionally, a Voluntary Fisheries Adjustment Scheme (VFAS) was completed in 2023 to buy out some WCDSIMF entitlement (fishing hours) and remove a further 38.5 t of demersal scalefish catch (revised recovery benchmark of 180 t).

There was a total of 60 WCDSIMF licences in 2023, of which less than half fished for demersal scalefish in that year (Table 5.1). The average length of WCDSIMF vessels is around 15 m, with typical fishing trips ranging from one to several days. Fishers in the WCDSIMF operate throughout the year, however, temporal closures are in place (for all fishing sectors) to protect spawning Snapper in Cockburn Sound and Warnbro Sound (1 August-31 January) and Baldchin groper in the Abrolhos Islands (1 October-31 December).

WCDSIMF catches of demersal scalefish are primarily retailed locally in WA, with some occasionally sold to east coast markets. The estimated gross value of production (GVP) for the WCDSIMF in 2023 was \$1-5 million, with more than 100 people employed in jobs relating to this industry.

Table 5.1. Summary of key attributes of the commercial WCDSIMF.

Attribute	
Fishing methods	Handline, dropline
Fishing capacity	Kalbarri: 5,595 hrs (2023) Mid-West: 9,711 hrs (2023) South-West: 3,803 hrs (2023)
Number of licences	60 (2023)
Number of vessels	28 active in 2023
Size of vessels	6.4 – 26.2 m in length
Number of people employed	>100
GVP of fishery	\$1 – 5 million

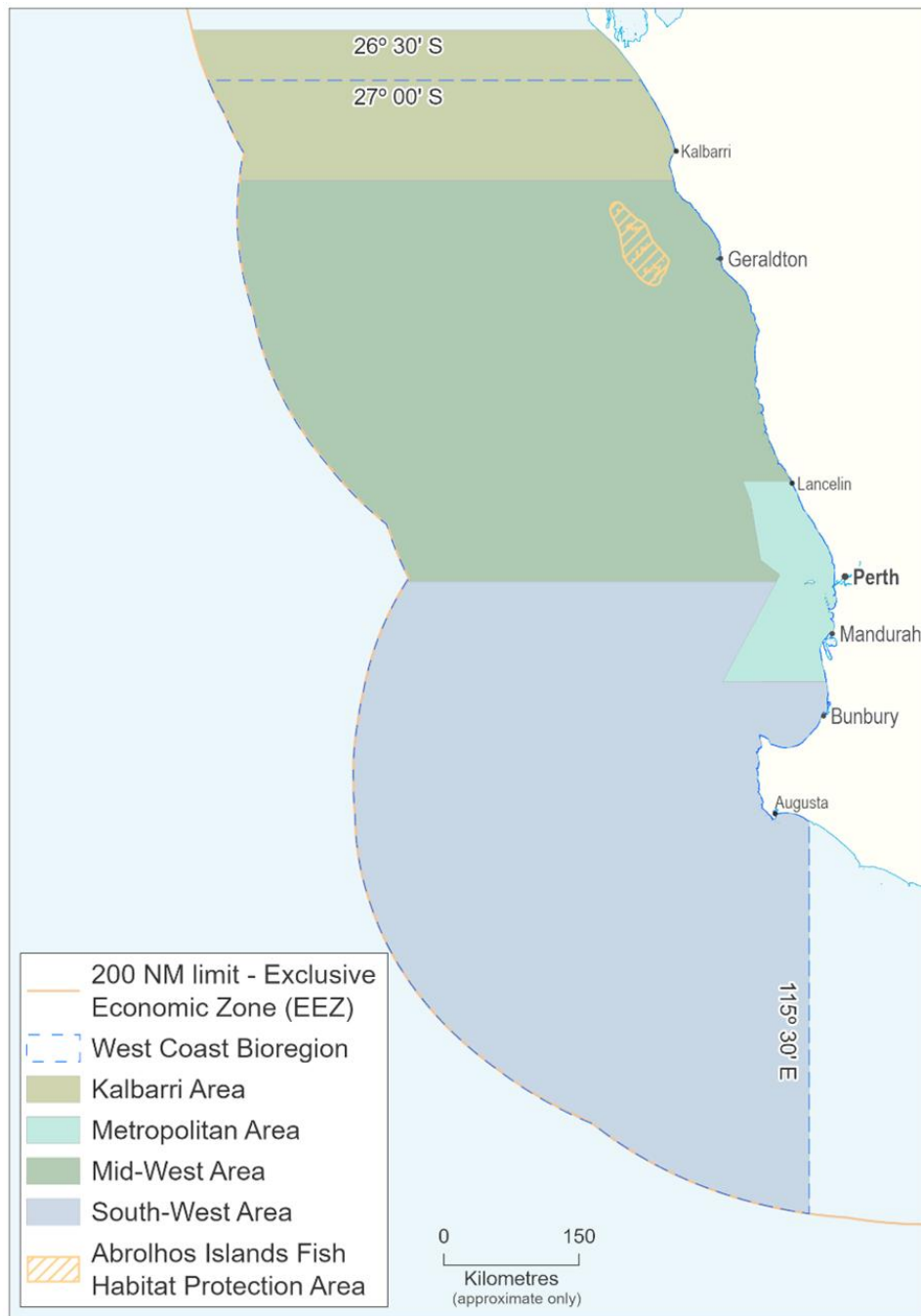


Figure 5.3. Boundaries of the commercial WCDSIMF and its management areas.

5.2.3 Fishing Methods and Gear

The WCDSIMF is a handline and dropline fishery, with operators permitted to retain all scalefish except species that are under specific State or Commonwealth management (e.g. mackerel and tunas). A handline refers to a fishing line which, when used for fishing, has either one end in constant contact with an authorised boat, is attached to a fishing rod or reel, or is held in the hand. Conversely, a dropline means a fishing line, excluding a longline, which, when used for fishing, is weighted at one end only, is deployed vertically through the water, and is not attached to a boat. Hooks are typically baited with scaly mackerel, pilchards, mullet and squid.

5.2.4 Susceptibility

The WCDSIMF targets demersal scalefish in depths between 20 and 250 m. With the exception of the Metropolitan area and other spatial closures (including marine park sanctuary zones), the fishery operates within most of the spatial range of the key target species. Fishing with handlines and droplines is highly selective for demersal scalefish and fishers specifically target areas of suitable habitat for these species. Although immature fish are rarely landed by WCDSIMF fishers, release rates vary both spatially and temporally in response to variation in the recruitment of key species. Trends in recent data suggest that the majority (>90%) of fish released by WCDSIMF fishers are Snapper.

5.3 Recreational and Charter Fisheries

5.3.1 History of Development

Recreational fishing has always been a popular past time in WA, particularly for iconic demersal scalefish species such as WA dhufish and Snapper. The estimated number of recreational fishers in WA has increased from 315,000 in 1989-90 (Lindner and McLeod 1991) to 620,000 in 2020-21 (DPIRD 2021). Line fishing, primarily from boats, is the main method used by recreational fishers to target demersal scalefish, with some spear fishing also occurring (Ryan et al. 2022). Charter boat operators have also been offering fishing tours to paying clients in WA for many years (Telfer 2010), providing opportunities for recreational fishers without their own boats to access this resource. Charter fishing has traditionally been managed as a component of the recreational fishing sector.

Historical data on recreational fishing in WA are limited. Early creel surveys undertaken across the WCB indicate that fishing effort increased substantially from the mid-1990s to mid-2000s (Lai et al. 2019). The close to 50% increase in the estimated hours fished by private recreational fishers in the WCB between 1996-97 and 2005-06 was likely related to human population growth, increased boat ownership and improved access to boating facilities during this time, particularly in the Metropolitan and South-West areas (Lai et al. 2019). Data from these early recreational creel surveys and charter logbooks introduced in 2001 suggest that annual demersal scalefish catches collectively retained by the recreational fishing sector in the WCB were close to 400 t in the mid-2000s (Figure 5.4).

Recreational fishing for demersal scalefish in WA has primarily been managed by minimum size and bag limits. In response to increasing concerns about declining catches and stock levels of key demersal species in the WCB (Wise et al. 2007), a suite of new management arrangements was introduced in 2010 to reduce recreational catches of demersal scalefish in this region by at least 50% of 2005-06 levels (i.e., below 250 t). These measures included reduced bag and boat limits for key demersal scalefish species, a 2-month temporal closure for demersal fishing by private recreational and charter fishers in the WCB, and the introduction of a Statewide fishing licence for boat-based private recreational fishers. The licence has provided a valuable sampling frame for recreational fishing surveys to allow monitoring of participation, effort and catch every 2-3 years in WA (e.g. Ryan et al. 2022).

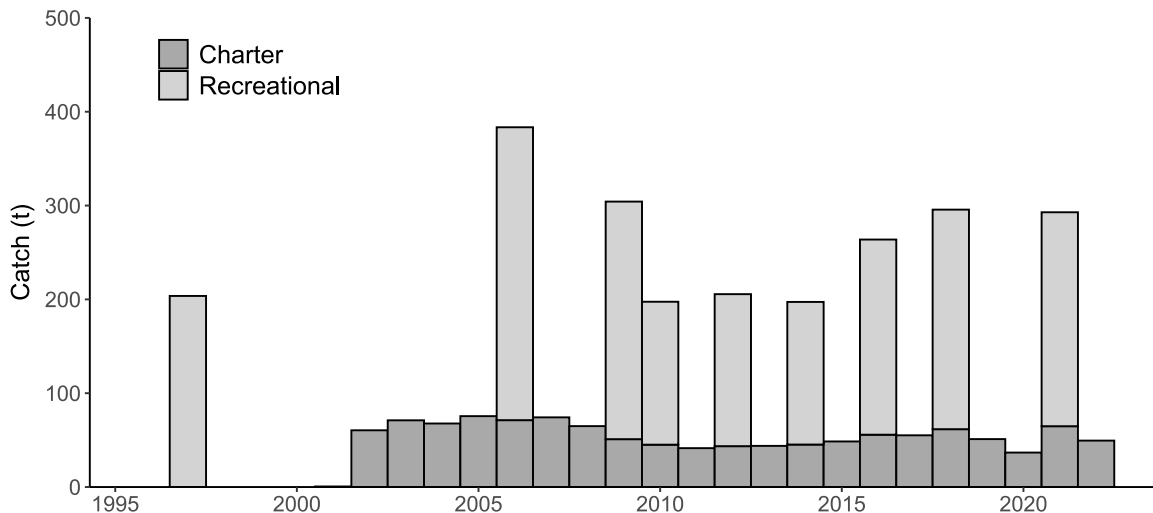


Figure 5.4. Annual retained catches (t) of demersal scalefish by boat-based charter and private recreational fishers in the WCB. These catches, which include the top-15 species/groups landed by these sectors, comprise charter catches reported in compulsory logbooks by financial year since 2001-02 (dark grey bars), and estimates of private boat-based recreational catches from fishing surveys undertaken since 1997-98 (light grey bars).

Following the implementation in 2010 of new management measures to recover the WCDSR, demersal scalefish catches by private recreational fishers in the WCB remained relatively steady at around 150 t between 2010-11 and 2013-14, before increasing to more than 200 t since 2015-16 (Figure 5.4). In contrast, charter catches of demersal scalefish in the WCB have remained relatively stable since the early 2000s, broadly fluctuating around 50-60 t annually (Figure 5.4). The charter catch decreased in 2019-20, due to impacts of COVID-19 restrictions on charter activities, before returning to pre-pandemic levels the following year (Figure 5.4). Research on COVID-19 impacts on private recreational fishing in WA showed no decline in vessel retrievals at Metropolitan boat ramps during the most stringent restrictions in 2020, with regional boat ramps to the north experiencing increases in vessel activity once travel restrictions eased (Ryan et al. 2021).

Information from the Statewide recreational fishing surveys undertaken since 2011-12 show that demersal scalefish catches by private boat-based recreational fishers in the WCB broadly follow the same pattern as changes in the overall fishing effort of this sector (Figure 5.5). The estimated number of boat days, fishing trips and hours fished by boat-based recreational fishers in the WCB declined between 2011-12 and 2013-14 but have since shown a steady increase back to their 2011-12 levels. While recognising that these trends reflect the overall recreational fishing effort in the WCB, including boat-based methods other than line fishing, they have largely been driven by changes in fishing activity in the Metropolitan area. In contrast, the much lower levels of recreational fishing effort in the Kalbarri/Mid-West and South-West areas have remained relatively stable over the last decade (Figure 5.5).

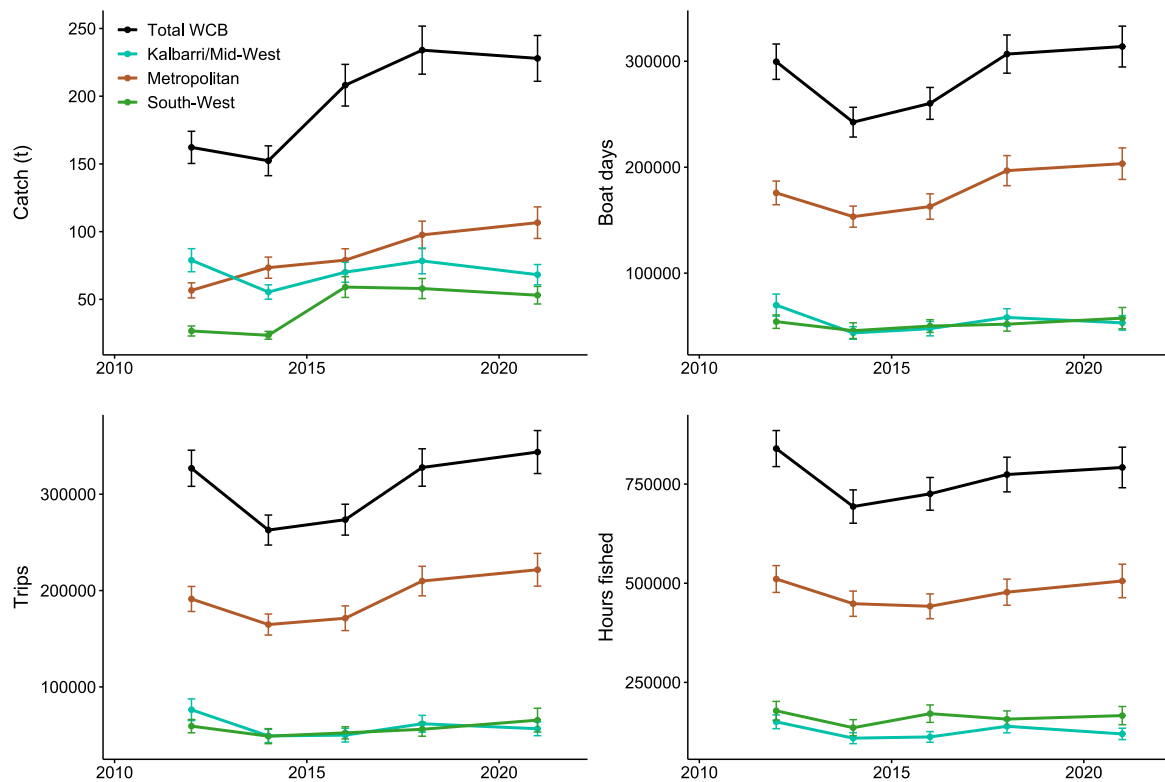


Figure 5.5. Periodic estimates (± 1 SE) of demersal scalefish catch (t), number of boat days, fishing trips and hours fished by private boat-based recreational fishers in the WCB since 2011-12. Note that effort measures include all recreational fishing in the WCB, including methods other than line and spearfishing.

The number of charter operators reporting catches of demersal scalefish in the WCB has decreased from more than 80 vessels in the early 2000s to around 50-60 vessels since the 2010 management changes (Figure 5.6). While charter effort in the early 2000s was primarily focused in the Metropolitan area, a declining number of active vessels in this area has resulted in the majority of recent charter activity in the WCB occurring in the Kalbarri/Mid-West (Figure 5.6). In the South-West, the annual number of active charter vessels and trips recording demersal scalefish catches have remained low, as is also reflected in much lower charter catches compared to those landed in other areas of the WCB (Figure 5.6).

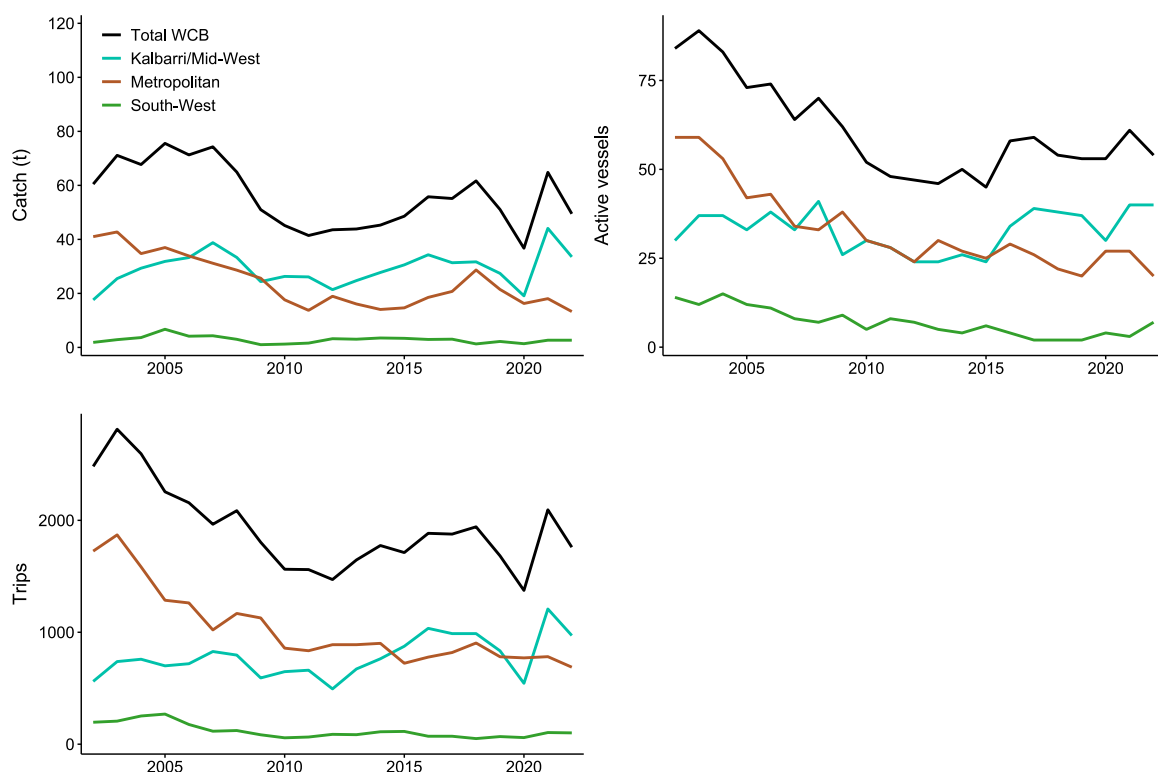


Figure 5.6. Annual demersal scalefish catch (t), number of active vessels and fishing trips by charter fishers in the WCB since 2001-02. Note that effort measures include all charter vessels reporting catches of demersal scalefish in the WCB, excluding fishing undertaken while diving.

5.3.2 Current Fishing Activities

Summaries of the key attributes of the recreational and charter fisheries for demersal scalefish in the WCB are provided in Table 5.2 and Table 5.3, respectively. The recreational fishing sector in the WCB can be broadly separated into three zones; Kalbarri/Mid-West (27°S–30°S), Metropolitan (30°S–32°S) and South-West (32°S–115°30'E). Recreational and charter fishing is prohibited in sanctuary zones of State and Commonwealth Marine Parks within the WCB. The majority of recreational fishing in the WCB occurs from the major population centres, as well as smaller regional towns that are popular destinations for travelling tourists and fishers, particularly during school holidays.

There were more than 100,000 boat-based recreational fishing licences issued to residents in the WCB in 2020-21 (Ryan et al. 2022), with around 40,000 of these estimated to have fished for demersal scalefish in the WCB at least once during that year. While there have been around 60 charter vessels recording demersal scalefish catches in recent years, new management arrangements implemented in 2023 for the charter sector limits access to the WCDSR to 33 operators that, based on their catch histories, have been allocated tags for landing key species (WA dhufish, Snapper, Baldchin groper and Breaksea cod). Vessels used by private recreational and charter fishers to target the WCDSR vary in size, with the majority of fishing events in this region being single-day trips.

In 2023, to achieve a further 50% reduction in the recreational recovery benchmark to 135 t (115 t private recreational and 20 t charter), the 2-month temporal closure was expanded to around half of the year, combined with bag and boat limit changes. In addition to the temporal closures in place to regulate private recreational fishing effort in the WCB, species-specific spawning closures (for all fishing sectors) are also used to protect key spawning locations for Snapper in Cockburn Sound and Warnbro Sound (1 August – 31 January) and Baldchin groper in the Abrolhos Islands (1 October – 31 December).

Recreational fishers in WA collectively contributed an estimated \$1.1 billion to the State's economy in 2018-19, supporting more than 9,000 jobs (Moore et al. 2023). While these estimates comprise all recreational fishing activities across WA, the economic impacts of recent management actions taken to support the recovery of the WCDSR are already being realised for many tackle shop and charter businesses.

Table 5.2. Summary of key attributes of the private boat-based recreational fishery for demersal scalefish in the WCB.

Attribute	
Fishing methods	Hook and line
Fishing capacity	Unrestricted
Number of licences	109,621 total (WCB, 2020-21)
Number of vessels	~40,000 targeting demersal scalefish (WCB, 2020-21)
Size of vessels	Varied
Number of people employed	9,380 (WA, 2018-19)
Value of fishery	\$1.1 billion (WA, 2018-19)

Table 5.3. Summary of key attributes of the charter fishery for demersal scalefish in the WCB.

Attribute	
Fishing methods	Hook and line
Fishing capacity	7,500 quota tags* (WCB, 2023-24)
Number of licences	~100 (WA)
Number of vessels	33 eligible for demersal tags (WCB)
Size of vessels	Varied
Number of people employed	Unknown
Value of fishery	Unknown

*Total allocation for WA dhufish, Snapper, Baldchin groper and Breaksea cod, with other species able to be landed subject to the charter vessel being in possession of tags (unless fish are caught and consumed on the vessel).

5.3.3 Fishing Methods and Gear

Recreational fishers that target demersal scalefish in the WCB are almost exclusively boat-based (Ryan et al. 2022), with some shore-based fishing for species like Snapper also occurring (Tate and Smallwood 2021; Tate et al. 2022). Fishers typically target this resource with rods and reels or handlines, using a variety of baits and lures. A small proportion of recreational and charter catch of demersal scalefish is also taken by spear fishers.

5.3.4 Susceptibility

The recreational fishing sector primarily operates out of major population centres and tourist towns with access points such as boat ramps. While the majority of fishing occurs on daytrips, a growth in ownership of larger boats and the willingness of fishers to travel greater distances to catch fish over a far wider area has led to an extension of the spatial extent of recreational fishing effort in the WCB (Wise et al. 2007). The fishing efficiency of recreational fishers has also increased as a result of the greater affordability of GPS and fish finders (Marriott et al. 2011). The recent removal of size limits for WA dhufish, Baldchin groper and Breaksea cod aims to minimise the number of fish being released and thus negative impacts of post-release mortality.

5.4 Customary Fishing

Fish resources in estuarine and nearshore waters have provided sustenance to Aboriginal people in south-western Australia for thousands of years. In a description of the cultural seascape of Wadandi Boodja in the south-west of WA, the connections of Traditional Owners to sea country are highlighted through several songlines named by key demersal fish species (Davies et al. 2022). For example, *Gortjguttuk Kwala* (the Pink Snapper songline) originates in *Waatu Waugal* water (Geographe Bay) and describes a movement of fish around the Cape Naturaliste into *Waarten Waugal* water (West Coast) in the hot and dry *Birak* (December-January) season.

Although data on customary fishing for demersal scalefish in the WCB is limited, catches of key species are likely negligible.

5.5 Illegal, Unreported or Unregulated Fishing

There are no estimates of illegal catches of demersal scalefish in the WCB. Fishers that do not follow the rules are subject to severe penalties.

6 Fisheries Management

6.1 Management Framework

The *Fish Resources Management Act 1994* (FRMA) provides the overarching legislative framework to implement the management arrangements for the WCDSR. The resource is currently managed under a 20-year recovery plan, with a combination of input and output controls used to maintain the retained catches and mortality of released fish (collectively referred to hereafter as total removals) of demersal scalefish below recovery benchmarks for the commercial and recreational fishing sectors.

As set out in the WCDSR harvest strategy (DPIRD 2021), recovery benchmarks are in place for the overall suite of demersal species in the WCB, as well as for key species and species groups that comprise the majority of catches. These 'allowable catches' were allocated to the commercial and recreational fishing sectors in line with formal resource sharing arrangements for the WCDSR (see DPIRD 2021 for more details). Once the resource has been recovered, a constant exploitation approach will resume where catches are expected to vary in proportion to variations in stock abundance.

Management arrangements for the WCDSR were updated in 2023 to help ensure key indicator species can rebuild to sustainable levels by 2030. Following a review of the 2021 WCDSR assessment outcomes against the recovery milestones specified in the harvest strategy (DPIRD 2021), the Minister for Fisheries approved a 50% reduction in the original recovery benchmarks for each sector. The 20-year recovery timeframe for the WCDSR is consistent with globally accepted standards to rebuild stocks of long-lived species (MSC 2022), for which extended periods at low stock levels increases the risk of recruitment impairment.

The commercial WCDSIMF is currently effort-managed, with each management area allocated a maximum number of effort units that can be used on an annual basis. Units are allocated to licence holders and provide entitlement in hours of fishing time, noting the Metropolitan area is allocated zero hours. Since the 2023 VFAS, the total capacity of the fishery is set at a level intended to maintain the total removals of demersal species below a new recovery benchmark for the fishery of 180 t (220 t total for the commercial sector). The use of a satellite-based Vessel Monitoring System (VMS) allows fishing effort in the WCDSIMF to be closely monitored and entitlement use acquitted accordingly. As described in the *West Coast Demersal Scalefish (Interim) Management Plan 2007*, licence holders also need to comply with a number of gear and other restrictions, including maximum numbers of lines and hooks that can be used and arrangements around transporting fish.

Since 2010, all persons fishing from a powered boat anywhere in the State are required to hold a Recreational Fishing from Boat Licence (RFBL), or fish in the company of a licence holder. Temporal closures have been identified as the primary measure used to maintain the total removals of demersal scalefish by recreational fishers below the updated recovery benchmark of 125 t. Recreational fishing for demersal scalefish in the WCB is now permitted for 6.5 months of the year, with three individual open seasons coinciding with key school holiday periods to reduce the impact of closures on tourism

and small businesses. Bag, boat and possession limits are also in place to limit recreational catches during the open seasons.

New management arrangements for the charter sector in the WCB, including a tag-based quota system, were implemented on 1 July 2023. To maintain catches below the updated 30 t recovery benchmark for the charter fishery, 33 Fishing Tour Operator Licence (FTOL) holders were allocated a total of 7,500 tags for any Snapper, WA dhufish, Baldchin groper and Breaksea cod landed in the WCB in 2023-24. All charter boats are still permitted to catch a limited number of demersal scalefish for the purpose of providing a meal for those on board the vessel as part of the fishing tour without the need to be in possession of tags. Annual reviews of logbook data will be used to determine the number of tags issued to eligible charter fishers each year.

In addition to the sector-specific management arrangements for the WCDSR, some resource-level measures are in place for all commercial, charter and recreational fishers. These supporting measures include size limits for species with moderate or low levels of post-release mortality (e.g., Snapper and Emperor spp.), as well as spawning closures for Snapper and Baldchin groper to protect specific areas where these species are known to become more easily targeted by fishers during their spawning seasons.

6.2 Harvest Strategy

Harvest strategies are formal decision-making frameworks that are widely recognised as essential tools for managing fisheries resources (e.g., Smith et al. 2014). They outline the overarching management objectives and provide a description of the performance indicators and reference levels used to measure how well the objectives are being met. A key component of harvest strategies is the control rule, which links the outcomes of stock assessments with appropriate management responses, designed to maintain those indicators at specified target levels (Bessell-Browne et al. 2022).

A WCDSR harvest strategy was published in 2021 (DPIRD 2021). In line with the State and Commonwealth harvest strategy policies and guidelines (Department of Fisheries 2015; Sloan et al. 2014; Department of Agriculture and Water Resources 2018), the key management objectives for targeted species focus on maintaining the biomass and fishing mortality for stocks above and below their respective threshold reference levels, which correspond to the levels at which Maximum Sustainable Yield (MSY) can be achieved. The WCDSR harvest strategy also includes details of the formal recovery plan developed to monitor the recovery of the resource against specific recovery milestones (DPIRD 2021).

The WCDSR harvest strategy was developed by a stakeholder-based reference group including representatives from DPIRD, the WA Fishing Industry Council (WAFIC), Marine Tourism WA and Recfishwest. This same group continues to be engaged to undertake the annual and periodic review processes outlined in the harvest strategy and provide recommendations to Government on the management of the resource.

6.3 External Influences

6.3.1 Environmental Factors

Recruitment success of many long-lived demersal scalefish species varies substantially between years and can be influenced in part by a range of environmental factors, including water temperature, productivity and the strength of ocean currents influencing the dispersal of eggs and larvae. Catches of species like Snapper and WA dhufish are typically dominated by a limited number of strong year classes, with the resilience of these stocks reliant on maintaining fishing pressure at a level where these cohorts can survive, grow and contribute to spawning for many years before being caught.

The sensitivity to climate change impacts on demersal scalefish in the WCB have been assessed as ranging from low to medium for key species (Newman et al. 2024). Specifically, changes in the frequency and intensity of El Niño Southern Oscillation (ENSO) events, decadal variability in the strength of the Leeuwin Current, and increases in water temperature have the potential to influence the distribution and biological characteristics of key species. For example, the 2010-11 marine heatwave in WA caused observed mortalities of demersal species in waters around the Abrolhos Islands (Pearce et al. 2011b) and a high recruitment of Baldchin groper at the most southern extent of its geographic range (Cure et al. 2018a).

6.3.2 Other Activities

Cockburn Sound, Warnbro Sound and Owen Anchorage are currently the only recognised locations in the WCB where snapper consistently form large aggregations to spawn. The eggs and larvae spawned in Cockburn Sound are retained by a gyre, with juveniles remaining within this nursery area until they move offshore at approximately 18 months of age (Wakefield et al. 2011). Cockburn Sound is a highly industrialised area and associated environmental degradation may affect the viability of these spawning aggregations including reductions in spawning biomass, egg and larval quality and growth of juveniles. Environmental impact assessments will be conducted in relation to the proposal to construct a new container ship port in Cockburn Sound, following a number of scientific studies to provide data on abundance, distribution and importance of Cockburn Sound for targeted fish species.

Fish known to produce and detect sounds of varying frequencies, including WA dhufish (Parsons et al. 2013), have the potential to be affected by underwater noise and seismic surveys within surrounding areas. Known effects of anthropogenic sounds on fish include mortality, physical injury, hearing impairment and changes in behaviour (Popper and Hawkins 2019), with the severity of impact likely to be influenced primarily by the proximity of the fish to the sound source. Environmental impact assessments are currently being progressed for a proposed seismic survey in part of the Mid-West area in 2025-26 and for a number of proposed offshore windfarm projects in the WCB.

7 Information and Monitoring

7.1 Range of Information

A range of information is available to support stock assessments and harvest strategy reviews for the WCDSR (see Table 7.1). This information has been collected primarily by fishery-dependent sampling of commercial, charter and recreational catches across the different management areas of the WCB. Limited fishery-independent sampling is also undertaken for juvenile fish, used to supplement the biological data sets from which estimates of life history parameters for the key species are derived.

Table 7.1. Summary of information available for assessments of the WCDSR.

Data type	Fishery-dependent/independent	Purpose/Use	Frequency of collection	History of collection
Commercial catch and effort statistics	Dependent	Monitoring of commercial catch and effort trends, catch per unit effort (CPUE)	Monthly returns	Since 1975
			Daily logbooks	Since 2006-07 (TDGDLF) Since 2008 (WCDSIMF)
Catch Disposal Records	Dependent	Monitoring of demersal catches in WCRLMF	By trip	Since 2009
VMS data	Dependent	Effort entitlement consumption in WCDSIMF	By trip	Since 2008
Charter catch and effort statistics	Dependent	Monitoring of charter catch and effort trends	Monthly (by trip)	Since 2001-02
Recreational catch and effort statistics	Dependent	Monitoring of recreational fisher catch and effort trends	Periodic annual surveys	1996-97 2005-06 2008-09 2009-10 2011-12 2013-14 2015-16 2017-18 2020-21
Biological data	Dependent and independent	Life history parameters, age and length compositions, inputs to stock assessments	Periodic (near annual for key species, opportunistic for others)	Since 1990s

7.2 Monitoring

7.2.1 Catch and Effort

Commercial fishing effort and retained catches of demersal scalefish in the WCB have been recorded in compulsory monthly returns since 1975 and, more recently, in daily logbooks for the key fisheries targeting the WCDSIMF. The daily logbooks, which record catches (in weight) and effort information at a finer spatial scale (10×10 nm blocks) compared to the monthly returns (60×60 nm blocks), have been in place for the TDGDLF since 2006-07 and the WCDSIMF since 2008. Demersal scalefish catches for personal consumption by operators in the WCRLMF are derived from compulsory Catch Disposal Records (CDRs). These data are combined for annual reporting of commercial WCDSR catches, and to derive annual catch time series for stock assessments of key species. VMS data are used to calculate use of annual effort entitlement in the WCDSIMF.

It has been a statutory requirement for charter tour operators to submit monthly returns detailing their daily fishing effort and catches since 2001-02. These logbooks provide data on the number and lengths of all retained fish, as well as the numbers of released fish. Information on fishing effort and (retained and released) catches of private recreational fishers in the WCB is derived from recreational fishing surveys undertaken periodically since the mid-1990s. Early creel surveys in 1996-97, 2005-06, 2008-09 and 2009-10 used an on-site bus-route methodology to collect information on boat-based fishing effort and catches (in numbers) through fisher interviews (Lai et al. 2019). As on-site surveys exclude access points such as marinas, and fishing that occurs outside of the survey times, catch estimates from creel surveys are adjusted to account for such 'out of scope' fishing in the WCB.

Since the introduction of the RFBL in 2010, a state-wide survey of boat-based private recreational fishing has been conducted every 2-3 years using the licence register as a sampling frame (Ryan et al. 2022). This integrated survey uses three complementary components, including off-site phone-diary surveys, on-site boat ramp surveys and remote camera monitoring. For each survey, around 3,000 participants are provided with a diary kit for recording information on the numbers of fish retained and released each trip, with regular telephone interviews by trained staff minimising the occurrence of response errors such as recall bias and species misidentification (Lai et al. 2021).

The FishCatchWA tool was launched in November 2023 to allow fishers to record retained and released catches of key demersal species for each trip in the WCB. The data from this app, including information on uptake and usage, will be assessed on an ongoing basis with the aim for the app to provide more timely catch information that can complement ongoing periodic fishing surveys.

While compliance data provide information on levels of non-compliance to different types of fishery regulations (e.g., licence requirements, gear restrictions, spatial closures, size limits), there are no current estimates of illegal catches of demersal scalefish in the WCB.

7.2.1.1 Discards

Fishers release fish for a number of reasons, including being below the MLL for retention, being in excess of catch or bag limits, being damaged by depredation, or being considered of limited value (e.g., Ryan et al. 2021). In some cases, changes to fishery management can lead to unintended consequences such as increased numbers of released fish (Crisafulli et al. 2023). For species where released fish do not all survive, this additional source of mortality can have a substantial impact on stock sustainability (Crisafulli et al. 2025).

Since 2019, voluntary reporting and limited observer monitoring onboard WCDSIMF vessels operating across the three management areas indicate that fish released by commercial line fishers in the WCB primarily comprise Snapper (DPIRD unpublished data). The release rates (in numbers) of this species vary between management areas and fluctuate over time in response to large variability in annual recruitment (Fairclough et al. 2021). Since the MLLs for WA dhufish, Baldchin groper and Breaksea cod were removed in 2023, available data suggest releases of these species by commercial WCDSIMF fishers are rare.

A comprehensive observer monitoring program was undertaken on commercial TDGDLF gillnetting vessels operating throughout south-western WA since 1993 (Braccini et al. 2022). Species composition data collected during these fishing trips show that 5.5% of overall catches (in numbers) comprised discarded teleost (ray-finned) fishes, of which the majority were Buffalo bream and Dusky morwong.

Information on release rates (in numbers) by boat-based charter and private recreational fisheries are available from charter logbooks and periodic recreational fishing surveys, respectively. While release rates vary among species, data suggest 70-80% of overall Snapper catches and 40-50% of WA dhufish catches, were being released by charter and recreational fishers (Fairclough et al. 2021; Ryan et al. 2022).

Given the high release rates of key WCDSR species, the annual catch review process outlined in the harvest strategy for this resource explicitly accounts for all removals of fish by commercial, charter and recreational fishers (DPIRD 2021). The total removals of key demersal fish species, including both retained catches and estimates of post-release mortality of released fish, are compared to sector-specific recovery benchmarks in place to recover the resource (e.g., Fisher et al. 2023).

The process used to derive estimates of post-release mortality for key demersal species is outlined in the WDCSR harvest strategy (DPIRD 2021). For each species, it is calculated by multiplying the released catch (converted from numbers to weight based on available data) with the proportion of fish expected to die following release (e.g., 25% of Snapper, 50% of WA dhufish and Breaksea cod, and 90% of Baldchin groper; DPIRD 2021). Ongoing monitoring will evaluate how recent changes to WCDSR management, including the removal of MLLs for WA dhufish, Baldchin groper and Breaksea cod, have affected the release rates and associated post-release mortality by commercial, charter and recreational fishers.

7.2.2 Fishery-Dependent Monitoring

Biological data for key demersal scalefish species in the WCB have been periodically collected from line-based recreational and commercial catches since the early 2000s, with data from earlier biological studies of some species also available. These data typically include length, age and reproductive characteristics for each sampled fish, including their sex and stage of gonadal maturation (e.g., immature, resting, spawning) as determined by macroscopic (and microscopic) examination of the ovaries and testes. In some cases, the gonads of female spawning fish are also retained for histological and fecundity analyses. Tissue samples have also been collected from sampled fish to inform genetic studies on the stock structure of key species.

For both of the two key inshore indicator species (Snapper and WA dhufish), fishery-dependent monitoring of biological information since the early 2000s has been undertaken throughout the WCB near annually, with more recent sampling efforts focused on two consecutive years within each 3-yearly assessment cycle. For other species, biological samples have been obtained more opportunistically over this 20-year sampling period and may be limited to a specific area or fishing sector. Generally, biological sampling is undertaken across broad spatial and temporal scales to ensure the data are as representative as possible of the populations from which they were collected.

Biological samples collected from commercial catches include all management areas except for the Metropolitan area (as this is closed to commercial fishing for demersal species) and are typically obtained through fish processors. Since 2010, samples from the recreational sector have been derived mostly through the Send Us Your Skeletons (SUYS) program, which encourages recreational fishers to donate their fish frames to research (Fairclough et al. 2014a). With thousands of fishers donating their fish frames to date, the data derived from this program extend across the whole WCB and have been collected throughout the year when the fishery is open.

7.2.3 Fishery-Independent Monitoring

Fishery-independent monitoring of demersal scalefish in the WCB has been limited to date. Trawl surveys have been undertaken biannually (in autumn and late spring) in Cockburn Sound since 2008 to monitor blue swimmer crabs, however, they also collect data on the numbers and lengths of juvenile snapper (e.g., Johnston et al. 2007). Since 2008, Baited Underwater Remote Video (BRUV) surveys in Cockburn Sound and the nearby Warnbro Sound and Owen Anchorage have been conducted annually during winter to monitor abundance of juvenile Snapper. As length data are only available from BRUV surveys since 2021, it is only possible to distinguish between the different juvenile cohorts (year classes) in these very recent years. Analyses to evaluate the value of these data to develop an index of annual recruitment of Snapper are currently being undertaken.

A two-year project is currently underway to evaluate different fishery-independent survey methods for collecting biological samples and abundance information for WCDSR indicator species. This trial is undertaking seasonal surveys across different locations and depths in the Mid-West management area, using Remote Operated Vehicle (ROV), BRUV and line fishing methods.

7.3 Resource Assessments

The recovery of the WCDSR is currently monitored through annual reviews of available catch information for the demersal suite and key species/groups, as well as periodic assessments of stock status for each indicator species. This approach has been reviewed and agreed to by a stakeholder-based reference group as part of the development of the WCDSR harvest strategy (DPIRD 2021). The harvest strategy document includes details of the performance indicators and reference levels used to evaluate stock status, with clear and explicit recovery milestones and associated actions used to track the rebuilding process.

7.3.1 Annual Catch Review

The annual catch review process for the WCDSR considers estimates of the total removals of species in the WCB, comprising both retained catches and estimates of post-release mortality. Total removals are compared annually to recovery benchmarks for each sector, which are set to maintain fishing pressure at a level where the resource is expected to recover within an appropriate rebuilding timeframe. Annual catch reviews for the WCDSR are reported in the Department's annual Status Reports of the Fisheries and Aquatic Resources of Western Australia (State of the Fisheries; e.g., Newman et al. 2024).

7.3.2 Stock Assessments

Stock assessments of key indicator species are currently undertaken every 3 years to evaluate key performance indicators for stock status, including estimates of fishing mortality (F) and spawning stock biomass relative to unfished levels (B_{rel} ; Fairclough et al. 2021). Estimates of F and B_{rel} are compared to harvest strategy (target, threshold and limit) reference levels to monitor the rate of stock rebuilding against specific milestones, which have been agreed to by all sectors (DPIRD 2021).

Benchmark assessments of stocks comprise a range of different analyses and apply models of varying complexity to provide an overall understanding of stock status and the associated levels of risk, including:

- Level 1 analyses of trends in commercial, recreational and charter catches over time, relative to model predictions for the overall Maximum Sustainable Yield (MSY).
- Level 2/4 empirical and/or model-based assessments of trends in catch and catch per unit effort (CPUE, from fishery-dependent or fishery-independent sources) to evaluate changes in stock abundance.
- Level 3 empirical and/or model-based (catch curve and per recruit) analyses applied to size and/or age composition data and other biological information to provide estimates of F and B_{rel} .
- Level 5 model-based (integrated model) assessments applied to all available information (typically catch, CPUE, age and/or length data, and other biological data) to provide estimates of F and B_{rel} .

The levels of assessment undertaken varies among stocks (Table 7.2), with the relevance and applicability of each assessment level dependent on data availability. For example, Level 2 biomass dynamic models have been fitted to catch and CPUE information for Snapper and WA dhufish stocks but were deemed to provide uncertain outputs due to lack of signal in available CPUE indices (Fairclough et al. 2021). Irrespective of the assessment methodologies used, a weight-of-evidence approach is applied to consider all available information for a species, evaluating the outputs from each analysis to determine the current risk to stocks (Fletcher 2015).

Table 7.2. Levels of current assessment for stocks of key WCDSR species.

Stock	Level 1 (Catch)	Level 3 (Age-based)	Level 5 (Integrated)
Snapper (North)	×	×	×
Snapper (South)	×	×	×
WA dhufish	×	×	×
Baldchin groper	×	×	
Redthroat emperor	×	×	
Bight redfish	×	×	
Breaksea cod	×	×	
Hapuku	×		
Bass groper	×		
Blue-eye trevalla	×		

Several of the methods applied to assess key species of the WCDSR have been independently peer-reviewed by Haddon (in Wise et al. 2007) and O'Neill (2009; Table 7.3) and were deemed appropriate based on availability of information at those times. Since the first formal stock assessment of key WCDSR species was completed in 2007 (Wise et al. 2007), there have been substantial refinements in the approaches used to assess stocks (see Table 7.4). For example, post-release mortality of released fish is now explicitly accounted for within Level 3 and Level 5 assessment models (Fairclough et al. 2021; Crisafulli et al. 2025). Additionally, improvements in biological monitoring (e.g., the SUYS program; Fairclough et al. 2014a) and longer time series of fisheries catch and CPUE data have enabled the progression of Snapper and WA dhufish assessments from equilibrium Level 3 models to the development of more sophisticated and dynamic Level 5 models (Fairclough et al. 2021). Since 2021, the application of integrated assessment models to assess these species have also allowed for model projections to be explored under different scenarios of future catches (Fairclough et al. 2021). Although such projections are inherently uncertain, they provide valuable predictions of the level of catches expected to recover stocks.

Table 7.3. Main recommendations of O'Neill (2009) and actions completed to date.

Recommendation	Actions to date
Complete estimates of recreational catch and effort at least every two years	Integrated survey of boat-based recreational catch in WA has been undertaken every 2-3 years since 2011-12 (e.g. Ryan et al. 2022).
Maintain annual-time series of age composition data to estimate F	Almost annual fisheries-dependent monitoring of Snapper, WA dhufish and Baldchin groper (Mid-West) has been undertaken since 2002, with sampling of other key species undertaken opportunistically.
Develop age-structured population models to explore stock status and degree of overfishing	Integrated, age-structured assessment models have been implemented for stocks of Snapper and WA dhufish, for which CPUE data provide indices of abundance.
Explore use of longitudinal (cohort) methods to estimate improved measure of F	Catch curve methods that consider variable abundances of different cohorts are routinely applied for providing improved measures of F (relative to methods not allowing for recruitment variation).
Verify the appropriate shape of fishing selectivity schedules and allow modification due to changes in size limits	Per-recruit and integrated assessment models incorporate separate logistic gear selectivity and retention curves to allow for PRM of undersized fish, noting removal of size limits for WA dhufish, Baldchin groper and Breaksea cod since 2023. Age-based integrated models allow for changes in size limits that have occurred for this species over time.
Further develop annual time series of fishery-independent abundance indices for WA dhufish, Snapper and Baldchin groper	- Fishery-independent surveys to monitor adult relative abundance of key indicator species are currently being trialled in the Mid-West using ROV, BRUV and line fishing methods. - Trawl and BRUV surveys of juvenile Snapper have been undertaken annually in Cockburn Sound, with analyses to explore their value as an index of recruitment currently ongoing. - Projects to investigate juvenile habitats of WA dhufish (Lewis et al. 2012; Strzelecki et al. 2013) and Baldchin groper (Cure et al. 2015) completed. Substantial time periods of monitoring would be required to determine if these would reflect cohort relative abundance.
Develop formal quantitative management evaluation models	A single-species Management Strategy Evaluation (MSE) model was developed by Fisher et al. (2011) to allow the effectiveness of alternative management options (e.g. changes to bag limits, size limits, reductions in exploitation etc.) for stocks with different biological characteristics.
Develop the potential for commercial and recreational catch rates as indices of abundance	Standardised time series of commercial dropline and handline CPUE, adjusted for assumed changes in fishing efficiency over time, are now used as input to the integrated assessment models for Snapper and WA dhufish. (Recreational) charter CPUE have also been examined and presented for these two species (Fairclough et al. 2021).

Table 7.4. The evolution of stock assessments undertaken for WCDSR species.

Year completed	Species assessed	Methods used	Key findings
2007 ¹	Snapper, WA dhufish, Baldchin groper	Due to limited historical data, assessments were based on: • Equilibrium catch curve analyses of age composition data (most samples from 2003-06, pooled across years) to estimate long-term average F. • Simple, age-based per-recruit analyses using available biological information and assumed knife-edged selectivity to calculate yield per recruit (YPR) and eggs per recruit (EPR).	Estimates of F well above M , indicative stocks experiencing overfishing.
2014 ²	Snapper, WA dhufish, Baldchin groper	Assessments applied a number of refined methods, including: • Improved equilibrium catch curve analyses, fitted simultaneously to multiple consecutive years of age composition data (samples up to 2011) to account for recruitment variability, to estimate long-term average F. • Conventional, age-based per-recruit analyses using updated biological information and estimates of logistic selectivity-at-age, to estimate spawning potential ratio (SPR).	Slight decrease in F and increase in SPR for dhufish and snapper in the southern WCB, with continued high fishing pressure in northern WCB.
2017 ³	Snapper, WA dhufish	Assessments for the two key species were based on: • Improved equilibrium catch curve analyses, fitted simultaneously to multiple years of age composition data (samples up to 2015) to account for recruitment variability, to estimate long-term average F. • Preliminary evaluation of changes in F using a novel catch curve method accounting for a change in exploitation resulting from management reform. • Age-based per-recruit analyses, explicitly accounting for PRM (of fish below their MLL) and incorporating a stock-recruitment relationship (assumed steepness of 0.75) to account for the impact of fishing on recruitment.	Snapper and WA dhufish assessed as High risk, with outputs consistent with the prediction that it would take in order of 20 years to recover resource.
2021 ⁴	Snapper, WA dhufish	Included the first dynamic, integrated stock assessment models developed for these stocks: • Based on available time series of catch and commercial CPUE data (up to 2020), and age composition data sampled up to 2018. • Including two areas, fitted separately to data collected in the south (South-west and Metropolitan) and north (Mid-west and Kalbarri) management areas. • Using updated biological information, accounting for temporal changes in growth of WA dhufish. • Accounting for PRM of fish below MLL, based on scenarios with additional catch or modelled using a retention function approach (Shertzer et al. 2021).	Snapper and WA dhufish assessed as Severe and High risk, respectively, with model outputs (including projections based on average constant recruitment to stocks) suggesting that further management action would be required to recover resource by 2030.

¹ Wise et al. 2007 (updated in 2009 using age data up to 2008)

² Fairclough et al. 2014b

³ Department of Fisheries unpublished report (2017); see also Fairclough and Walters 2019

⁴ Fairclough et al. 2021

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Appendix 1. Productivity Susceptibility Analysis

Productivity Susceptibility Analysis

Productivity Susceptibility Analysis (PSA) is a semi-quantitative risk analysis originally developed for use in Marine Stewardship Council (MSC) assessments to score data-deficient stocks (Hobday et al. 2011; MSC 2022). While limited in its usefulness for providing stock status advice (i.e. as this approach does not provide estimates of fishing mortality or biomass), the PSA approach has been applied to produce a measure of the vulnerability of each key WCDSR species.

The PSA approach assumes that the overall sustainability risk for a stock depends on two characteristics: (1) the productivity of the species, which determines the capacity of the stock to recover if it is depleted, and (2) the extent of fishing impacts on the stock, determined by the susceptibility of the species to fishing activities. Scoring criteria used to assess the productivity and susceptibility of species to fishing are shown in Tables A1.1 and A1.2.

Productivity

Based on the life history characteristics of the key WCDSIMF species, their overall (average) productivity scores range from 1.57 for the moderately long-lived Baldchin groper, Redthroat emperor and Breaksea cod, to 2.00 for Snapper and the three offshore indicator species (Table A1.3). While also long-lived, the greater overall productivity scores (and lower risk scores) for WA dhufish and Bight redfish results from these species on average attaining maturity at a smaller size (< 400 mm). As broadcast spawning species with high potential annual fecundity, risk scores for the reproductive strategy and fecundity categories were low, however, their trophic levels were all scored as high risk (Table A1.3).

Susceptibility

As the key WCDSR species targeted by the commercial and recreational line sectors in the WCB, their susceptibility to fishing is scored high. The areal overlap of fishing effort with the spatial distribution of each inshore species was typically considered above 30%, while offshore species range into deeper waters that are not accessed by line fishers (Table A1.4). The vertical overlap is high, with sectors using gear specifically designed to target these species in their demersal habitats (Table A1.4). While line catches of immature individuals are generally rare (i.e., low selectivity scores), release rates of small Snapper can be high when strong new recruitment enter the fishery. As these species are typically retained by fishers rather than represent accidental bycatch species, post-capture mortality scores are high (Table A1.4).

Total PSA Risk

The overall PSA (vulnerability) risk scores ranged from High for Snapper, through Medium for WA dhufish and Baldchin groper, to Low for the remainder of the species (Figure A1.1).

Table A1.1. Productivity scoring table.

Productivity attribute	High productivity Low risk Score = 1	Medium productivity Medium risk Score = 2	Low productivity High risk Score = 3)
Average maximum age	<10 years	10-25 years	>25 years
Average age at maturity	<5 years	5-15 years	>15 years
Average maximum size (not to be used when scoring invertebrates)	<1000 mm	1000-3000 mm	>3000 mm
Average size at maturity (not to be used when scoring invertebrates)	<400 mm	400-2000 mm	>2000 mm
Reproductive strategy	Broadcast spawner	Demersal egg layer	Live bearer
Fecundity	>20,000 eggs per year	100-20,000 eggs per year	<100 eggs per year
Trophic level	<2.75	2.75-3.25	>3.25
Density dependence (only to be used when scoring invertebrates)	Compensatory dynamics at low population size demonstrated or likely	No depensatory or compensatory dynamics demonstrated or likely	Depensatory dynamics at low population sizes (Allele effects) demonstrated or likely

Table A1.2. Susceptibility scoring table.

Susceptibility attribute	Low susceptibility Low risk Score = 1	Medium susceptibility Medium risk Score = 2	High susceptibility High risk Score = 3)
Areal overlap (availability) i.e. overlap of fishing effort with stock distribution	<10% overlap	10-30% overlap	>30% overlap
Encounterability i.e. the position of the species / stock within the water column / habitat relative to the position of the fishing gear	Low encounterability / overlap with fishing gear	Medium overlap with fishing gear	High encounterability / overlap with fishing gear (Default score for target species in a fishery)
Selectivity of gear type Potential of gear to retain species	a) Individual < size at maturity are rarely caught	a) Individual < size at maturity are regularly caught	a) Individual < size at maturity are frequently caught
	b) Individual < size can escape or avoid gear	b) Individual < half the size can escape or avoid gear	b) Individual < half the size are retained by gear
Post-capture mortality i.e. the chance that, if captures, a species would be released and that it would be in a condition permitting subsequent survival	Evidence of majority released post-capture and survival	Evidence of some released post-capture and survival	Retained species or majority dead when released

Table A1.3. PSA productivity scores for each key species, with risk scores ranging from 1 (high productivity) to 3 (low productivity).

Species	Maximum age	Age at maturity	Maximum size	Size at maturity	Reproductive strategy	Fecundity	Trophic level
Snapper	3	2	2	2	1	1	3
WA dhufish	3	1	2	1	1	1	3
Baldchin groper	3	1	1	1	1	1	3
Redthroat emperor	3	1	1	1	1	1	3
Bight redfish	3	2	1	1	1	1	3
Breaksea cod	2	2	1	1	1	1	3
Hapuku	3	2	2	2	1	1	3
Bass groper	3	2	2	2	1	1	3
Blue-eye trevalla	3	2	2	2	1	1	3

Table A1.4. PSA susceptibility scores for sector that impact on each key species, with risk scores ranging from 1 (low susceptibility) to 3 (high susceptibility).

Species	Commercial				Recreational			
	Areal overlap	Vertical overlap	Selectivity	PRM	Areal overlap	Vertical overlap	Selectivity	PRM
Snapper	3	3	3	3	3	3	3	3
WA dhufish	3	3	1	3	3	3	1	3
Baldchin groper	3	3	1	3	3	3	1	3
Redthroat emperor	3	3	1	3	3	3	1	3
Bight redfish	3	3	1	3	3	3	1	3
Breaksea cod	3	3	1	3	3	3	1	3
Hapuku	2	3	1	3	2	3	1	3
Bass groper	2	3	1	3	2	3	1	3
Blue-eye trevalla	2	3	1	3	2	3	1	3

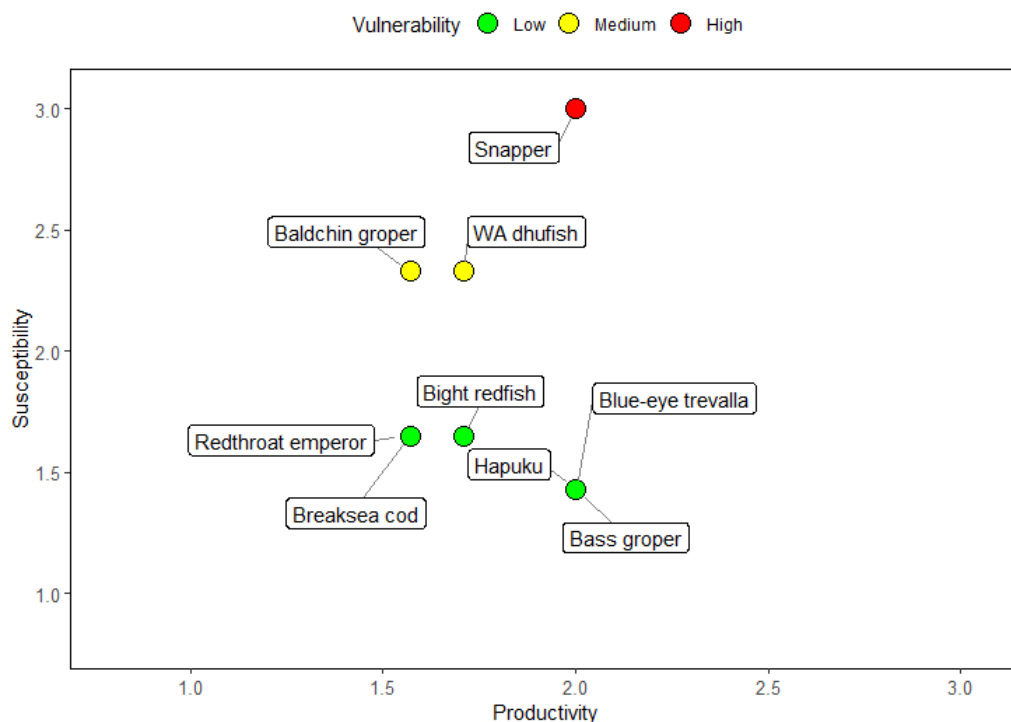


Figure A1.1. Productivity Susceptibility Analysis (PSA) risk scores for the key WCDSR species, noting risk scores for each category range from 1 (low risk) to 3 (high risk).

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